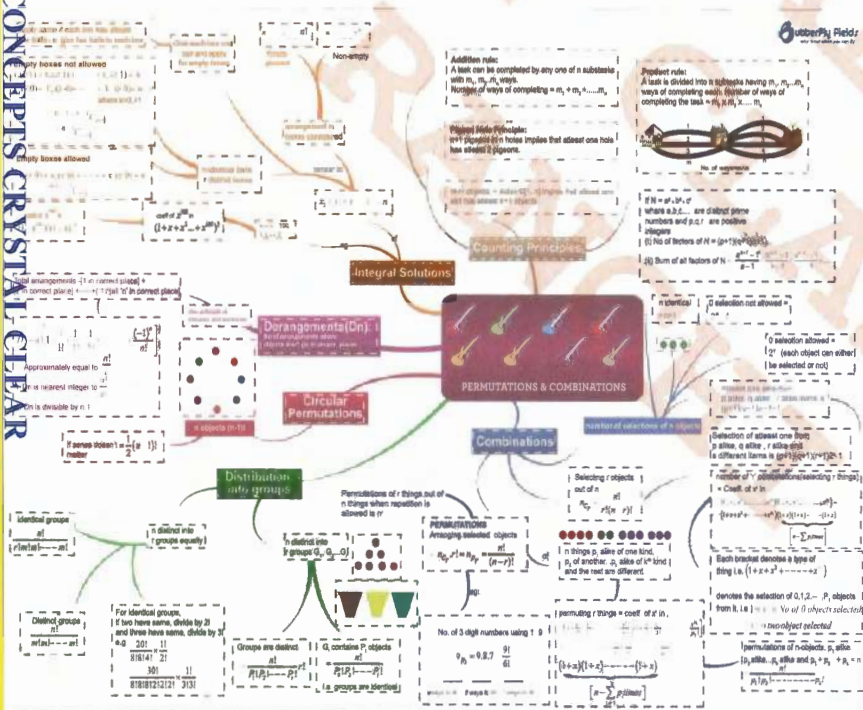


What is a Concept Map?

A Concept map is a colourful, visual form of representing key concepts covered in a chapter. It can be used as an effective overview, learning and revision tool.

At its heart is a central topic of the chapter. This is then explored by means of branches representing main ideas, which all connect to this central idea.

Concept maps are metacognitive tools that give students awareness towards the process of learning and allow the representation of knowledge structures in thematic fields.



Why use a Concept Map?

Students:

Brain is the key to effective learning for a student and more effectively the student uses it, the more successful he/she will be. If the brain of the student is stimulated with correct thinking and effective learning tools, excellent results can be obtained. **Understanding lists, lines, sequences, letters and numbers** is the work of the **LEFT BRAIN**. While **interpreting colour, images, rhythms and spatial awareness** is the work of the **RIGHT BRAIN**. Through a Concept Map both Right & Left parts of the brain would be put into use.

As a learning strategy, concept mapping stimulate learners' **commitment and interest towards the concepts**, which is very important for better learning. Students can use it as tool for effective revision of not just the current class chapters but also the topics covered previously.

Teachers:

As a teaching support, concept maps are also helpful, as they permit the establishment of networks of concepts that originate from a central concept in a coherent, structured and integrated manner, **guiding the sequence of teaching**. Concept maps can also be used by teachers for **curriculum planning**. Using this instrument means giving a new meaning to education, as well as a new meaning to the concepts of teaching, learning and assessment of learning.

Each main topic has been transformed into a Map. For example, given below is the map for CONIC SECTIONS which includes an introduction to the map.

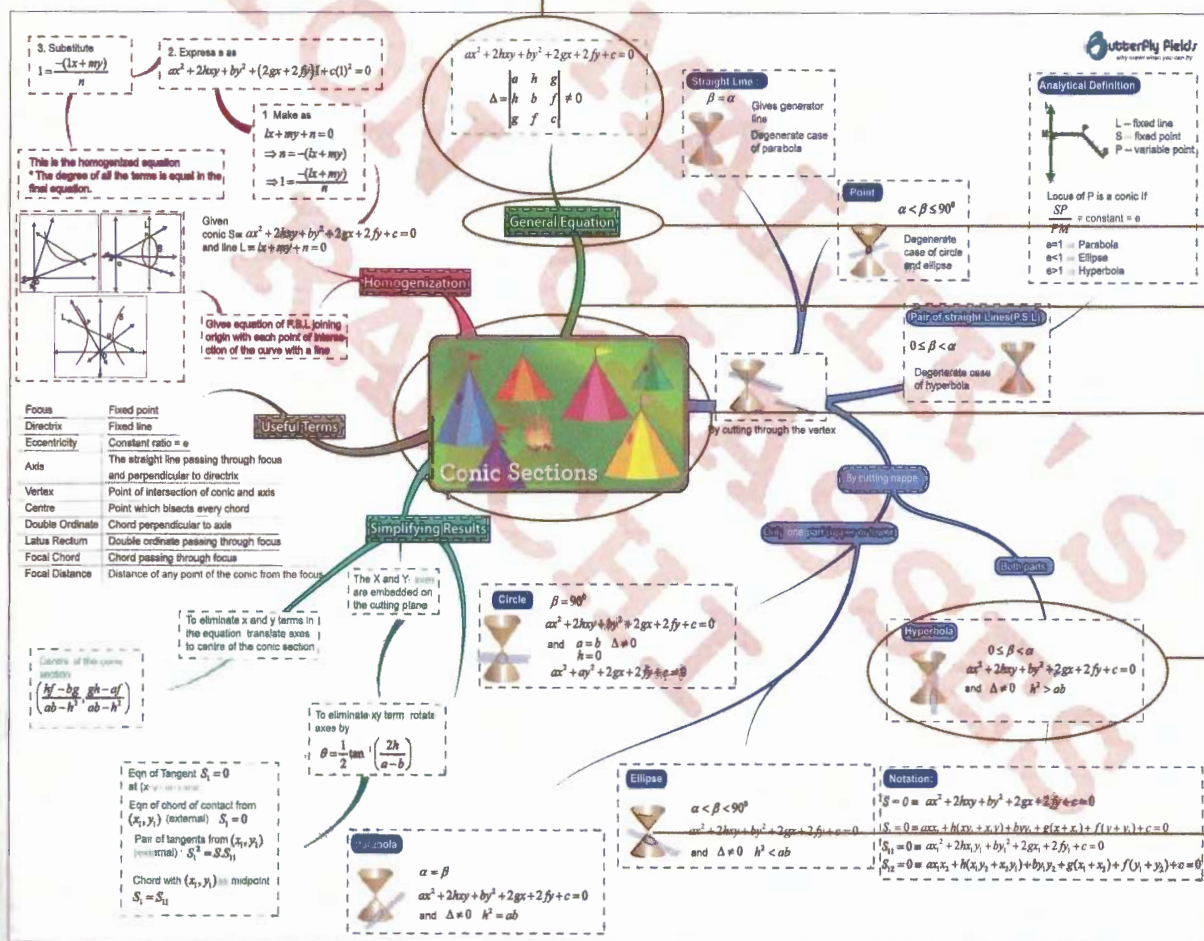
If you take the above branch as an example, we can understand that CONICS SECTIONS has a GENERAL EQUATION- $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

$$\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} \neq 0$$

Important Points

- The sub-concepts and the points under them are colour coded, i.e. each branch is identified with its colour.
- Based on the colour you can identify the sub-concept under which the point will come.
- Refer to the text book and identify the first branch to start with and revise one branch at a time.

Spend some time in understanding the Maps, once you get a hang of them they are a wonderful tool to get ahead in class.



The Sub-Concept

Branches - Each branch leads to a sub-concept of the chapter

The Central concept of the chapter

These are the important points under each sub / concept & these could be further branched out

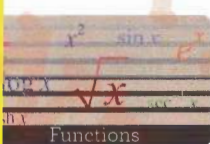
Image - Image is a mnemonic which will help you remember the concept

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A to Z of MATHS

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FUNCTIONS AND GRAPHICAL TRANSFORMATIONS

Topic	Page no.
Concept map	01,02
Questions	41
Solutions	53,54



LIMITS

Topic	Page no.
Concept map	09
Questions	43
Solutions	60,61



DEFINITE INTEGRATION

Topic	Page no.
Concept map	15
Questions	45
Solutions	66,67



PERMUTATIONS & COMBINATIONS

Topic	Page no.
Concept map	03
Questions	41
Solutions	54,55



DERIVATIVES

Topic	Page no.
Concept map	10
Questions	43
Solutions	61,62



RELATIONS

Topic	Page no.
Concept map	16
Questions	45
Solutions	67,68



BINOMIAL THEOREM

Topic	Page no.
Concept map	04
Questions	41,42
Solutions	55,56,57



APPLICATION OF DERIVATIVES

Topic	Page no.
Concept map	11
Questions	43
Solutions	62,63



PARABOLA

Topic	Page no.
Concept map	17
Questions	45
Solutions	68,69



STRAIGHT LINES

Topic	Page no.
Concept map	05
Questions	42
Solutions	57,58



CONTINUITY AND DIFFERENTIABILITY

Topic	Page no.
Concept map	12
Questions	44
Solutions	63,64



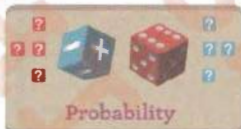
ELLIPSE

Topic	Page no.
Concept map	18
Questions	46
Solutions	69,70



CONIC SECTIONS

Topic	Page no.
Concept map	06
Questions	--
Solutions	--



PROBABILITY

Topic	Page no.
Concept map	13
Questions	44
Solutions	64,65



MATHEMATICAL INDUCTION

Topic	Page no.
Concept map	19
Questions	46
Solutions	70,71,72



CIRCLES

Topic	Page no.
Concept map	07,08
Questions	42,43
Solutions	59,60



SETS

Topic	Page no.
Concept map	14
Questions	44
Solutions	65,66



MATHEMATICAL LOGIC

Topic	Page no.
Concept map	20
Questions	46
Solutions	72,73

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A to Z of MATHS



IN EQUALITIES

Topic	Page no.
Concept map	21
Questions	47
Solutions	73,74



PROPERTIES & SOLUTIONS OF TRIANGLES

Topic	Page no.
Concept map	27
Questions	49
Solutions	79



RECTANGULAR CARTESIAN COORDINATES & LOCUS

Topic	Page no.
Concept map	35
Questions	51
Solutions	84,85



SEQUENCES AND SERIES

Topic	Page no.
Concept map	22
Questions	47
Solutions	74,75



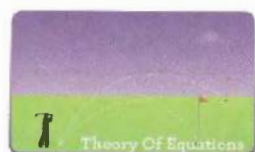
TRIGONOMETRY

Topic	Page no.
Concept map	28
Questions	49
Solutions	81



INTRODUCTION TO 3-D GEOMETRY

Topic	Page no.
Concept map	36
Questions	51
Solutions	85,86



THEORY OF EQUATIONS

Topic	Page no.
Concept map	23
Questions	47
Solutions	75,76



COMPLEX NUMBERS

Topic	Page no.
Concept map	29,30
Questions	49
Solutions	80



VECTORS

Topic	Page no.
Concept map	37,38
Questions	52
Solutions	88



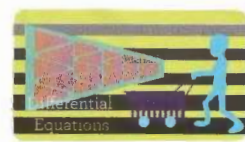
STATISTICS

Topic	Page no.
Concept map	24
Questions	48
Solutions	76,77



HYPERBOLA

Topic	Page no.
Concept map	31
Questions	50
Solutions	82



DIFFERENTIAL EQUATIONS

Topic	Page no.
Concept map	39
Questions	51
Solutions	86,87



EXPONENTIAL AND LOGARITHM SERIES

Topic	Page no.
Concept map	25
Questions	48
Solutions	77,78



MATRICES

Topic	Page no.
Concept map	32
Questions	50
Solutions	83



INDEFINITE INTEGRATION

Topic	Page no.
Concept map	40
Questions	52
Solutions	89,90



INVERSE TRIGONOMETRIC FUNCTIONS

Topic	Page no.
Concept map	26
Questions	48
Solutions	78



DETERMINANTS

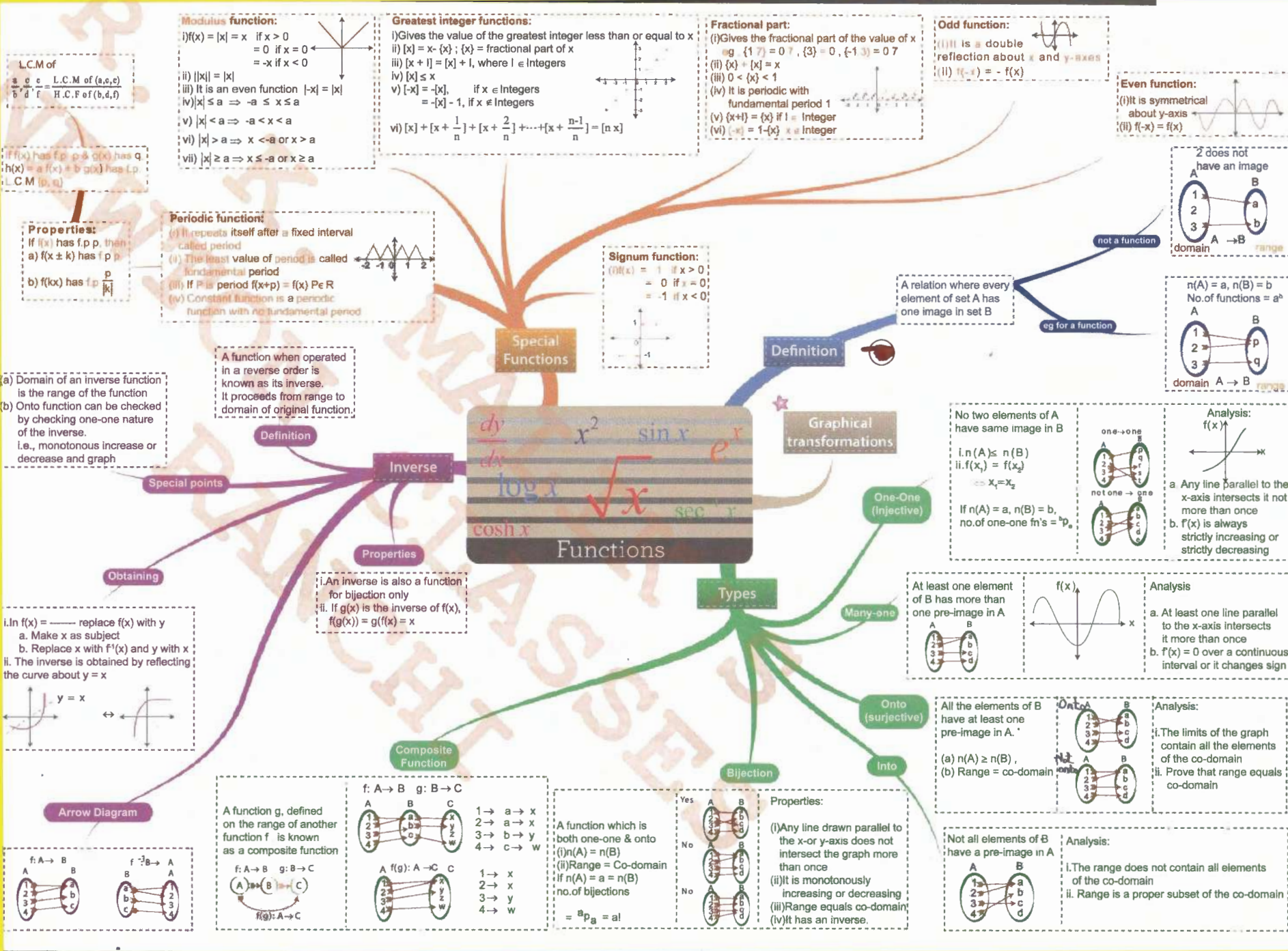
Topic	Page no.
Concept map	33,34
Questions	50
Solutions	83,84

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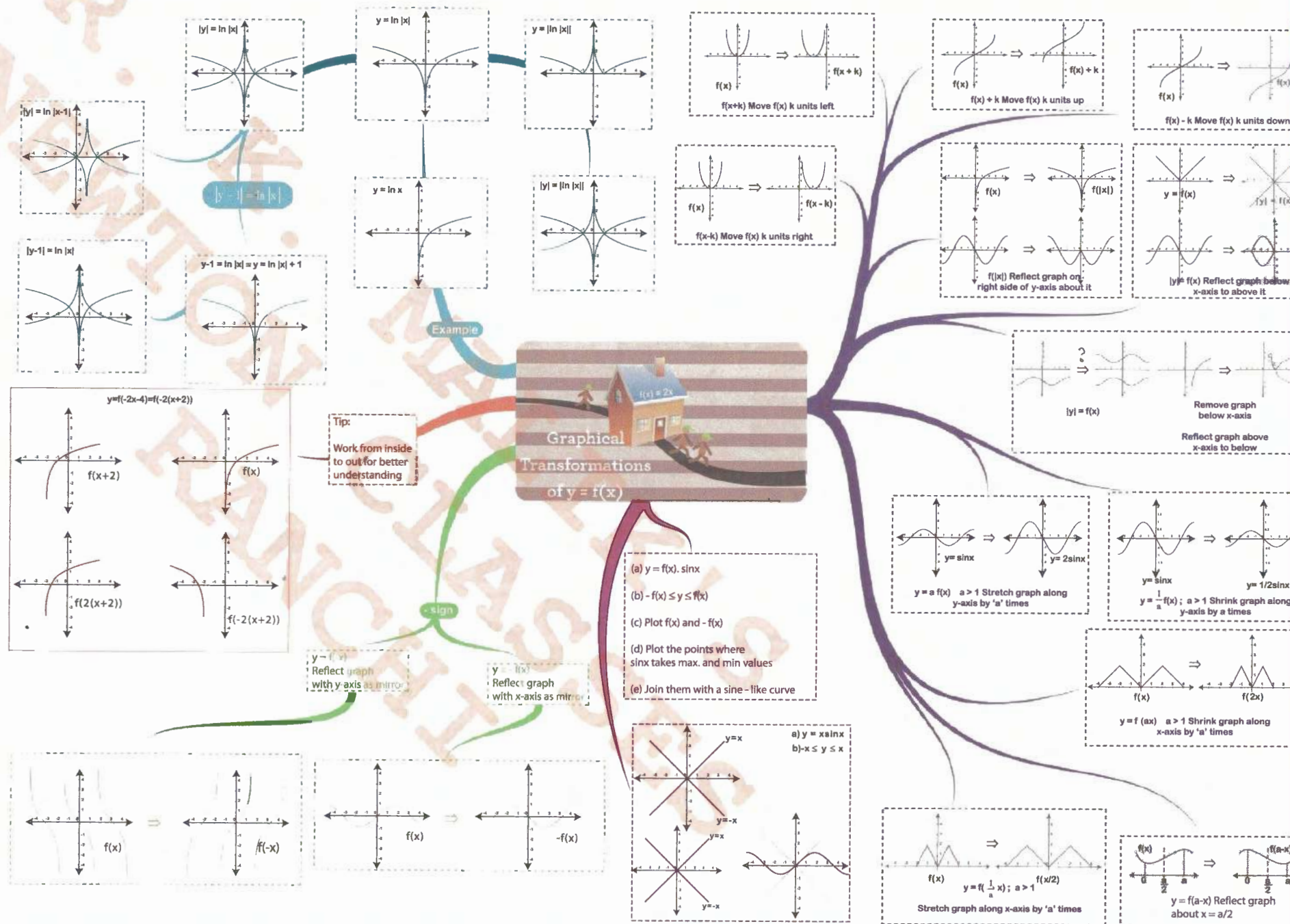
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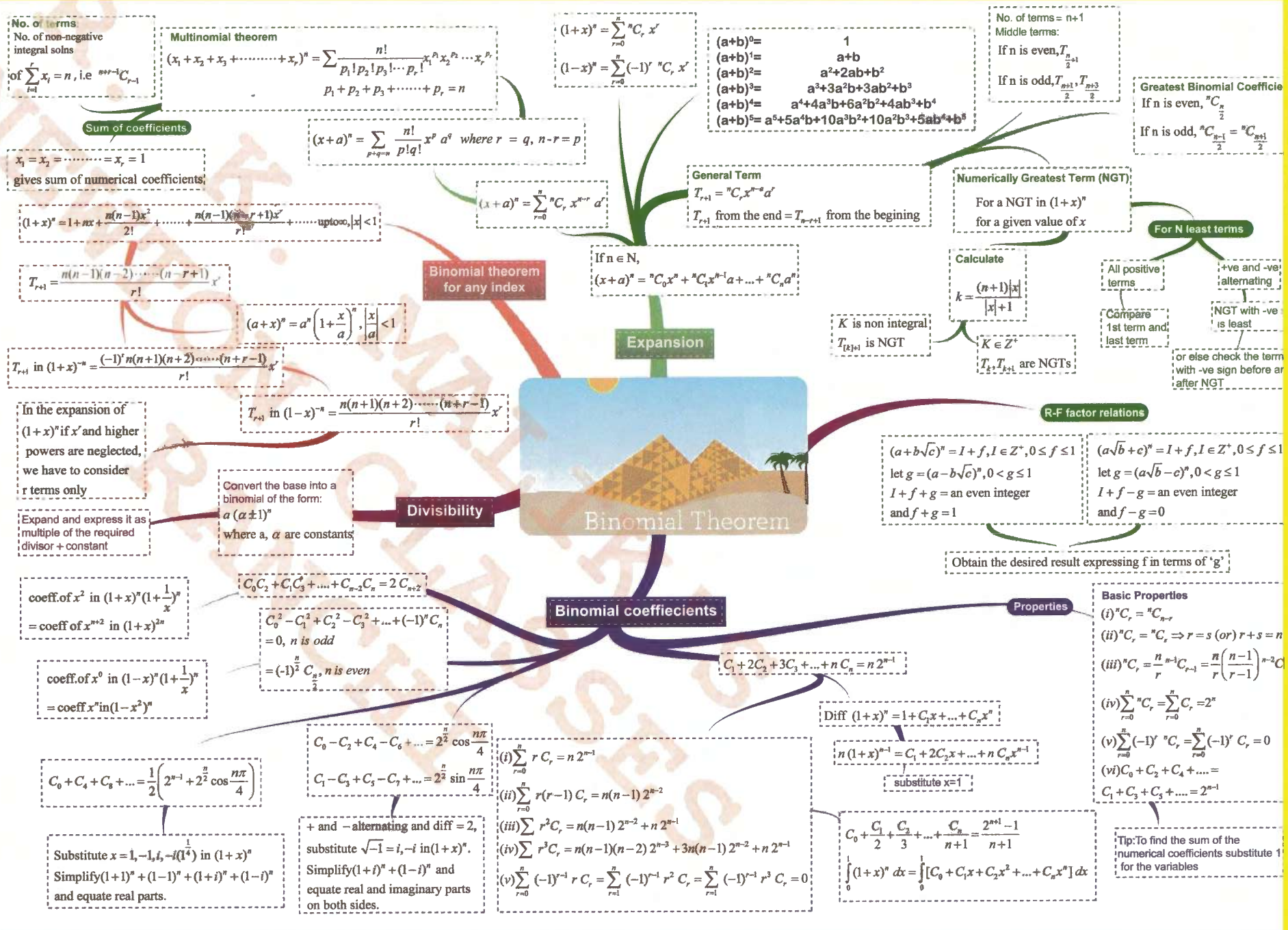
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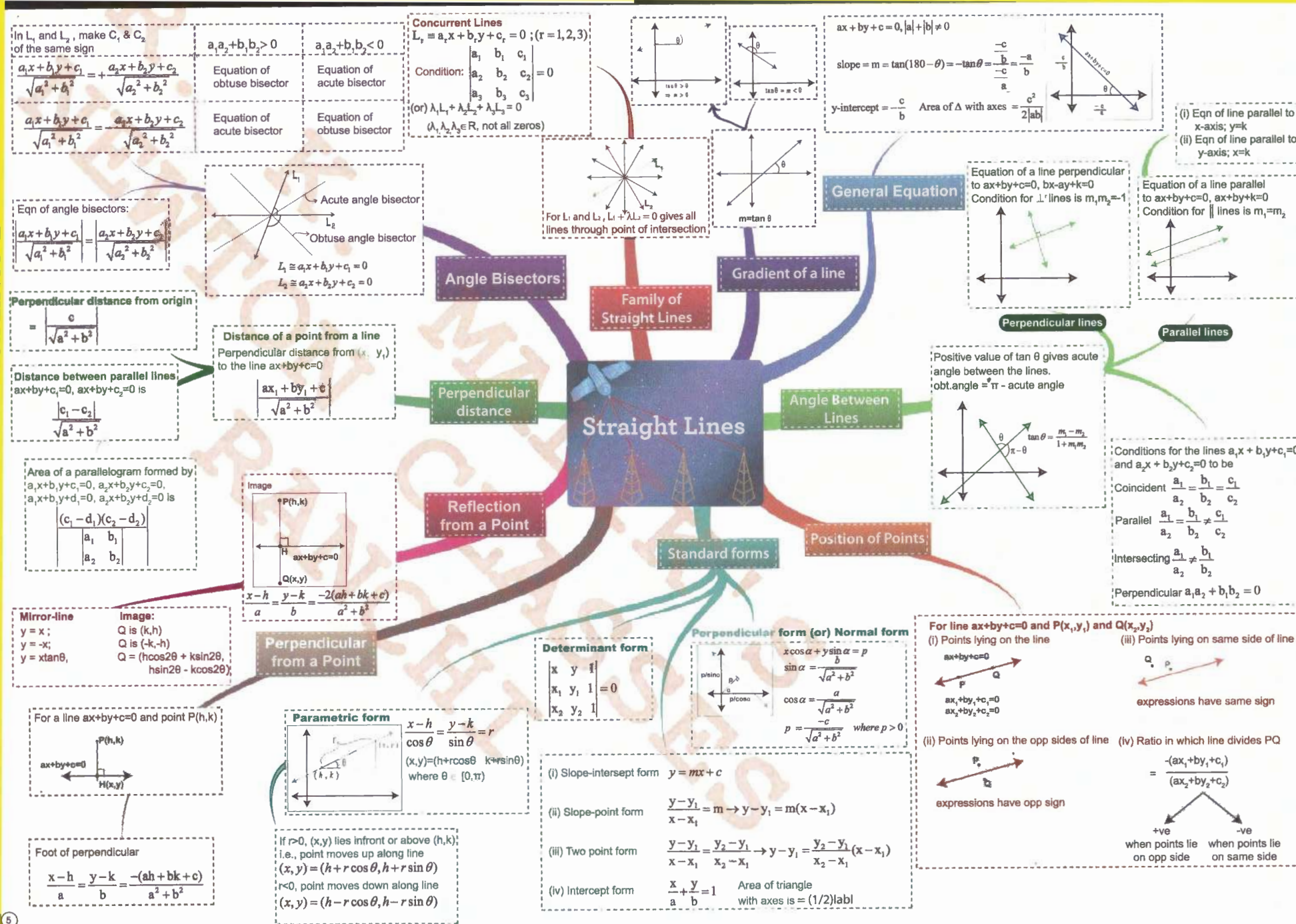
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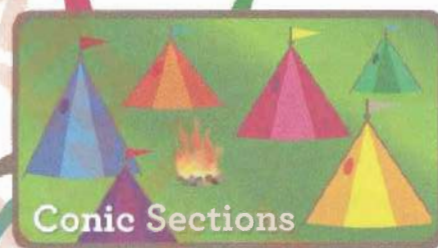
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Conic Sections

General Equation

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$$

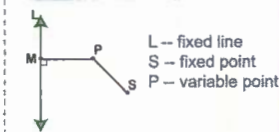
$$\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} \neq 0$$

Straight Line :

$$\beta = \alpha$$

Gives generator line
Degenerate case of parabola

Analytical Definition



Locus of P is a conic if

$$\frac{SP}{PM} = \text{constant} = e$$

$e=1 \Rightarrow$ Parabola
 $e<1 \Rightarrow$ Ellipse
 $e>1 \Rightarrow$ Hyperbola

Point

$$\alpha < \beta \leq 90^\circ$$



Degenerate case of circle and ellipse

Pair of straight Lines (P.S.L.)

$$0 \leq \beta < \alpha$$

Degenerate case of hyperbola



By cutting through the vertex

By cutting nappe

Only one part (upper or lower)

Both parts

Hyperbola



$$0 \leq \beta < \alpha$$

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$$

$$\text{and } \Delta \neq 0 \quad h^2 > ab$$

Ellipse



$$\alpha < \beta < 90^\circ$$

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$$

$$\text{and } \Delta \neq 0 \quad h^2 < ab$$

Notation:

$$S = 0 \equiv ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$$

$$S_1 = 0 \equiv axx_1 + h(xy_1 + x_1y) + byy_1 + g(x+x_1) + f(y+y_1) + c = 0$$

$$S_{11} = 0 \equiv ax_1^2 + 2hx_1y_1 + by_1^2 + 2gx_1 + 2fy_1 + c = 0$$

$$S_{12} = 0 \equiv ax_1x_2 + h(x_1y_2 + x_2y_1) + by_1y_2 + g(x_1+x_2) + f(y_1+y_2) + c = 0$$

Simplifying Results

The X and Y-axes are embedded on the cutting plane

Circle

$$\beta = 90^\circ$$

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$$

$$\text{and } a=b \quad \Delta \neq 0$$

$$h=0 \quad ax^2 + ay^2 + 2gx + 2fy + c = 0$$



To eliminate xy term, rotate axes by

$$\theta = \frac{1}{2} \tan^{-1} \left(\frac{2h}{a-b} \right)$$

Parabola



$$\alpha = \beta$$

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$$

$$\text{and } \Delta \neq 0 \quad h^2 = ab$$

Useful Terms

Focus	Fixed point
Directrix	Fixed line
Eccentricity	Constant ratio = e
Axis	The straight line passing through focus and perpendicular to directrix
Vertex	Point of intersection of conic and axis
Centre	Point which bisects every chord
Double Ordinate	Chord perpendicular to axis
Latus Rectum	Double ordinate passing through focus
Focal Chord	Chord passing through focus
Focal Distance	Distance of any point of the conic from the focus

To eliminate x and y terms in the equation, translate axes to centre of the conic section

$$\text{Centre of the conic section} \left(\frac{hf - bg}{ab - h^2}, \frac{gh - af}{ab - h^2} \right)$$

Eqn of Tangent $S_1 = 0$ at (x_1, y_1) on conic.

Eqn of chord of contact from (x_1, y_1) (external) $S_1 = 0$

Pair of tangents from (x_1, y_1) (external) $S_1^2 = S.S_{11}$

Chord with (x_1, y_1) as midpoint $S_1 = S_{11} = S_1$

3. Substitute $1 = \frac{-(lx + my)}{n}$

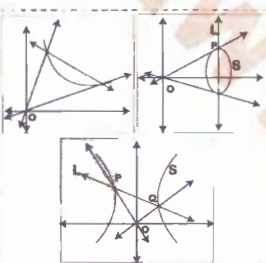
2. Express s as $ax^2 + 2hxy + by^2 + (2gx + 2fy)1 + c(1)^2 = 0$

1. Make as $lx + my + n = 0$

$$\Rightarrow n = -(lx + my)$$

$$\Rightarrow 1 = \frac{-(lx + my)}{n}$$

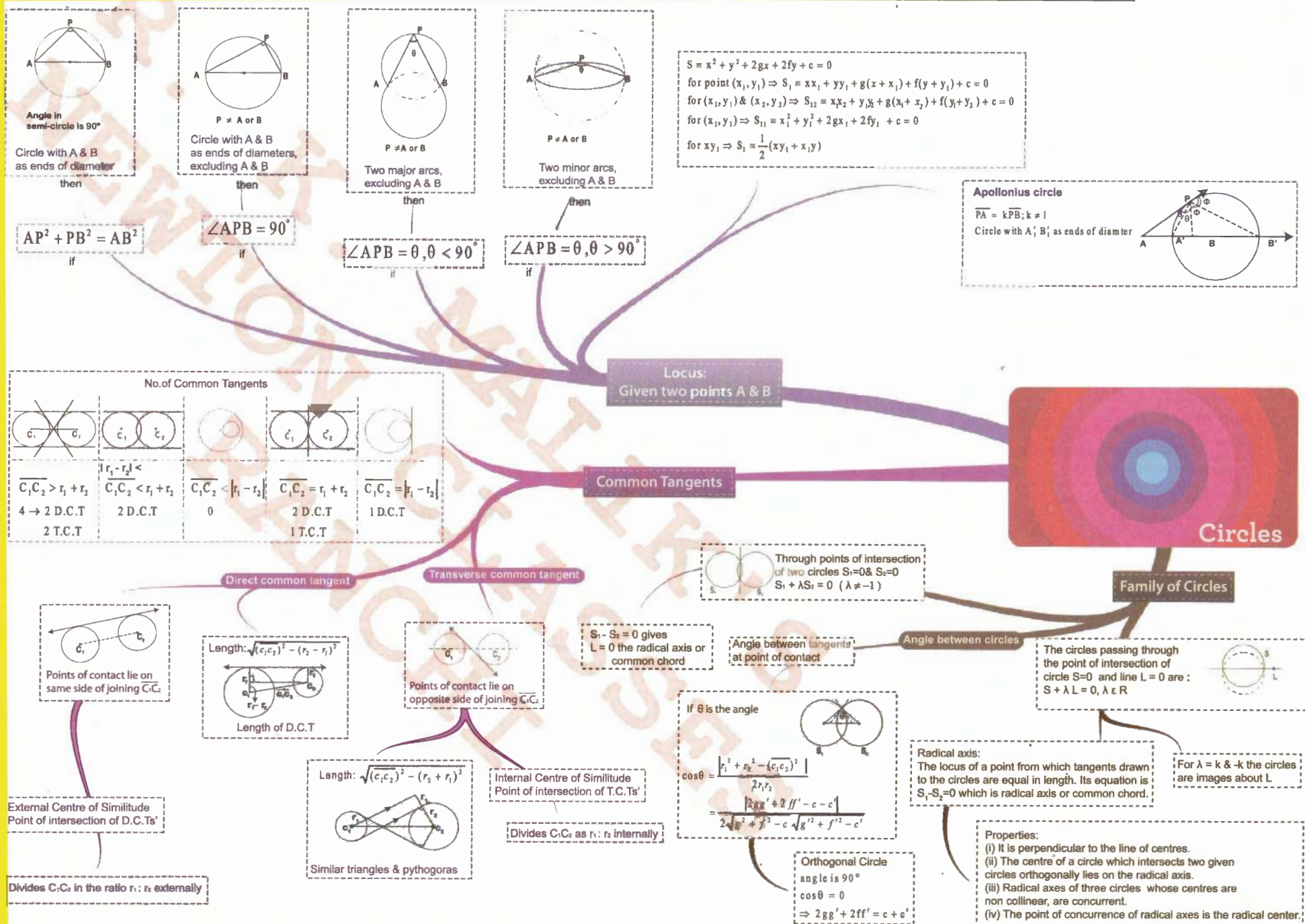
This is the homogenized equation
* The degree of all the terms is equal in the final equation.



Given conic $S \equiv ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$
and line $L \equiv lx + my + n = 0$

Gives equation of P.S.L joining origin with each point of intersection of the curve with a line

Homogenization

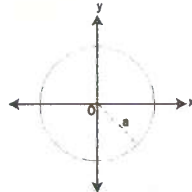


General Form
 $x^2 + y^2 + 2gx + 2fy + c = 0$ $c = (-g, -f)$ $r = \sqrt{g^2 + f^2 - c}$
 $(x - \alpha)^2 + (y - \beta)^2 = a^2$ $c = (\alpha, \beta)$ $r = a$

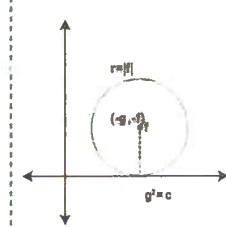
Equation of a circle with ends of diameter as (x_1, y_1) and (x_2, y_2)
 $(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$

Equation of a circle concentric with
 $x^2 + y^2 + 2gx + 2fy + c = 0$
 $x^2 + y^2 + 2gx + 2fy + k = 0$

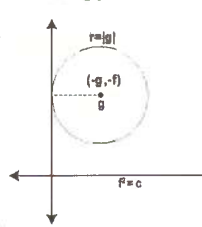
Simplest
 $x^2 + y^2 = a^2$; $c : (0, 0)$; $r = a$



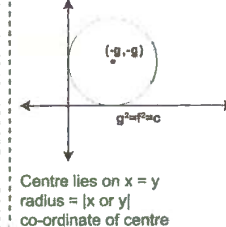
Touching x-axis



Touching y-axis



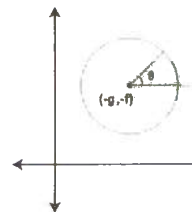
Touching x & y-axis



Passing through (x_1, y_1) , (x_2, y_2) , (x_3, y_3)

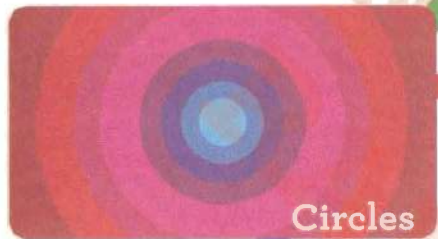
$$\begin{vmatrix} x^2 + y^2 & x & y & 1 \\ x_1^2 + y_1^2 & x_1 & y_1 & 1 \\ x_2^2 + y_2^2 & x_2 & y_2 & 1 \\ x_3^2 + y_3^2 & x_3 & y_3 & 1 \end{vmatrix} = 0$$

Parametric Form



$x^2 + y^2 + 2gx + 2fy + c = 0$
 Can be represented as
 $x = -g + r \cos \theta$
 $y = -f + r \sin \theta$
 $r = \sqrt{g^2 + f^2 - c}$

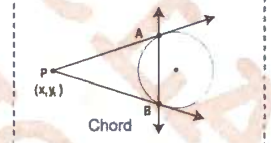
For simplest form $x^2 + y^2 = a^2$
 $x = a \cos \theta$, $y = a \sin \theta$



Equation

Properties

Chord of contact given by $S_1 = 0$



Length $\overline{PA} = \overline{PB} = \sqrt{S_{11}}$
 $\angle APC = \angle CPB = \frac{\theta}{2}$
 such that $\tan \frac{\theta}{2} = \frac{r}{\sqrt{S_{11}}}$

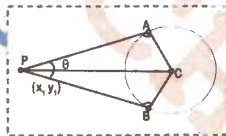
Chord whose mid-point is (x_1, y_1)
 $S_1 = S_{11}$

Tangent

A line that touches the circle at one point

Given by $S = 0$

$y = mx + c$ is tangent to $x^2 + y^2 = a^2$
 $\Rightarrow c^2 = a^2(1 + m^2)$



Equation of pair of tangents
 $S_1^2 = S \cdot S_{11}$

Quadrilateral PACB is concyclic if:
 (i) P, C are ends of diameter
 (ii) Centre lies on the circle

Normal:

Any line through centre perpendicular to tangent at point of contact with the circle

Eqn of a normal to $S = 0$ at (x_1, y_1) is
 $(x - x_1)/(x_1 + g) = (y - y_1)/(y_1 + f)$

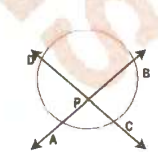
Properties

- Perpendicular to radius
- Perpendicular distance from centre to a line is 'd',
 $d < r \Rightarrow$ secant
 $d = r \Rightarrow$ tangent
 $d > r \Rightarrow$ doesn't intersect
- From an external point two tangents can be drawn to a circle
- The centre of a circle touching two lines, lies on the angular bisector of the two lines

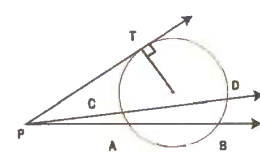


Director Circle:
 Locus of intersection of perpendicular tangents is a concentric circle with $\sqrt{2}$ times radius
 eg: Equation of director circle for $x^2 + y^2 = a^2$ is $x^2 + y^2 = 2a^2$

Secant Property:

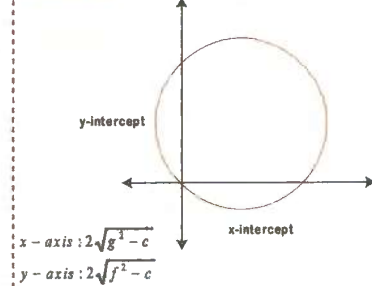


$\overline{PA} \cdot \overline{PB} = \overline{PC} \cdot \overline{PD}$



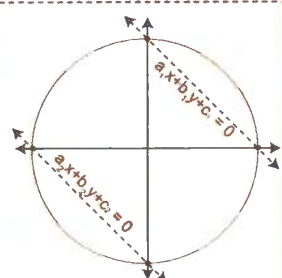
$\overline{PA} \cdot \overline{PB} = \overline{PC} \cdot \overline{PD} = \overline{PT}^2$

Intercept on:



x-axis: $2\sqrt{g^2 - c}$
 y-axis: $2\sqrt{f^2 - c}$

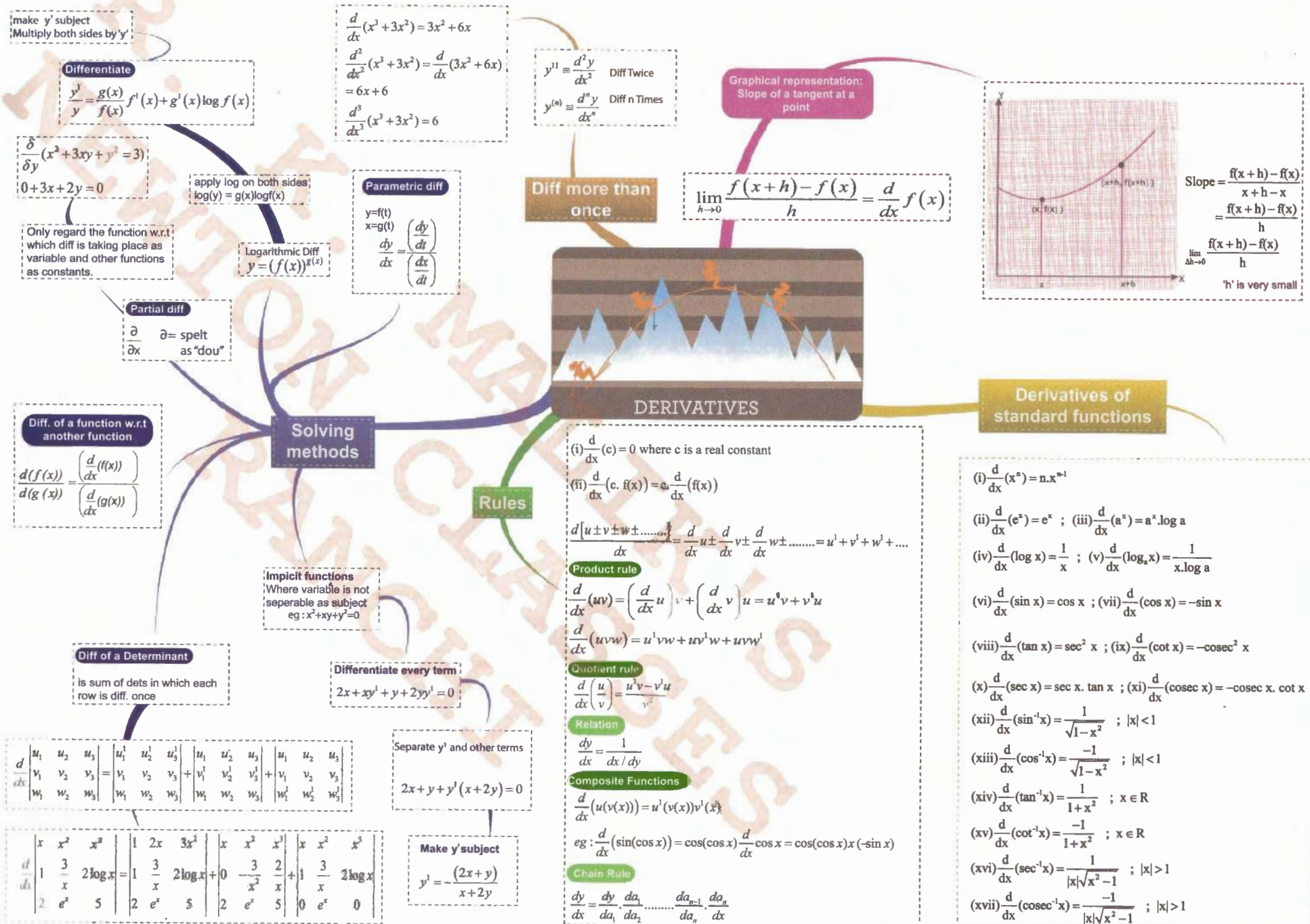
x-y intercepts of $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ are concyclic if $a_1a_2 = b_1b_2$

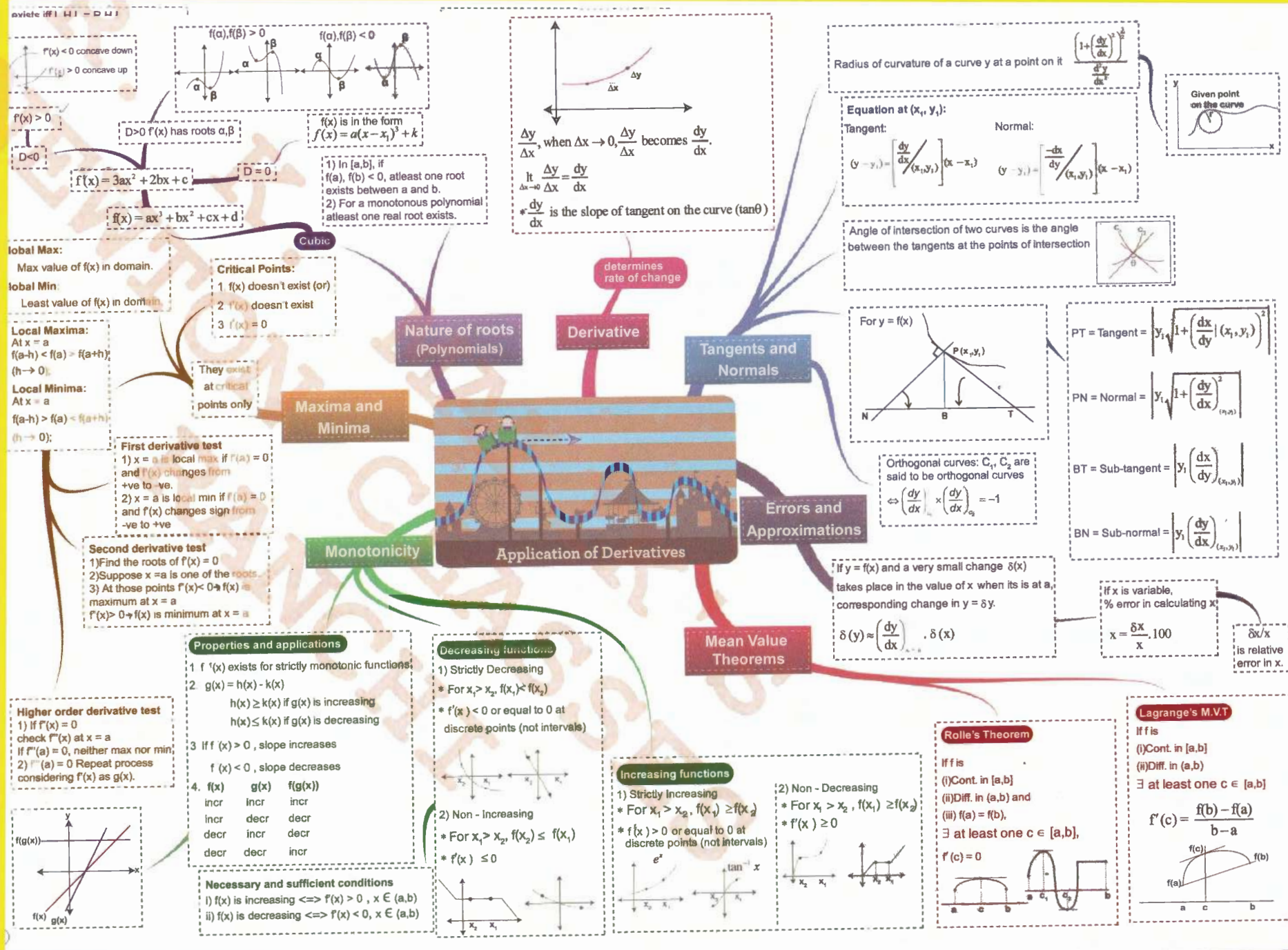


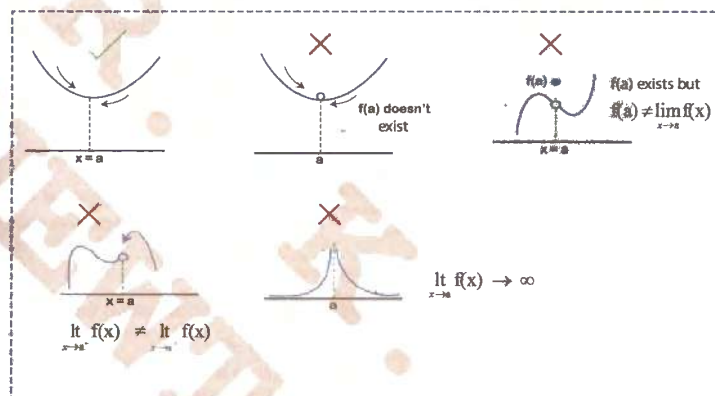
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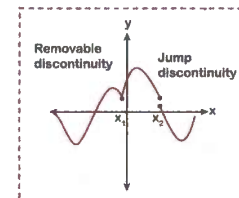






A function is continuous at $x=a$ if
 $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^-} f(x) = f(a)$

- If $f(x)$ and $g(x)$ are continuous at $x=a$
- $kf(x)$ is continuous at $x=a$
 - $f(x) \pm g(x)$ is continuous at $x=a$
 - $f(x)g(x)$ is continuous at $x=a$
 - $f(x)/g(x)$ is continuous at $x=a$ if $g(a) \neq 0$
 - If f is continuous on $[a,b]$ and $f: [a,b] \rightarrow [a,b]$ then $f(x) = x$ for some $x \in [a,b]$
 - $f(g(x))$ is continuous at $x=a$ and $f(x)$ at $g(a)$, $f(g(x))$ is continuous at $x=a$
 - If $g(x)$ is continuous at $x=a$ and $f(x)$ at $g(a)$, $f(g(x))$ is continuous at $x=a$
 - The points of disc. of $f(x)$ & $g(x)$ are not necessarily points of disc. of $f(g(x))$
eg: If $g(x) = -2$ $x \in \mathbb{R} \setminus \{0\}$
 $= 2$ $x \in \{0\}$ disc. at $x=0$
and $f(x) = x^2$
then $f(g(x)) = 4$ $x \in \mathbb{R}$
 - If $f(x)$ is cont. on $[a,b]$ such that $f(a)f(b) < 0$, then there exists atleast one solution of $f(x) = 0$ in (a,b)
 - If $f(x)$ is cont. on $[a,b]$ then it is bounded on $[a,b]$
ie $k \leq f(x) \leq l$ function some $k, l \in \mathbb{R}$



- Removable discontinuity: discontinuity at $x=a$
 $\lim_{x \rightarrow a} f(x) \neq f(a)$
- Discont. of 1st kind
 $\lim_{x \rightarrow a^-} f(x) \neq \lim_{x \rightarrow a^+} f(x)$
- Discont. of 2nd kind
neither L.H.L nor R.H.L exist

- A function isn't differentiable at sharp edges
 - A function is diff. if it is diff. at every point in the domain (open interval)
 - If f & g are diff, $f \pm g$, $f - g$ & f/g are diff.
(In f/g , not at points where g or $g' = 0$)
 - The composition of differentiable functions is differentiable.
 - Every where differentiable function:
diff. on $\mathbb{R}$, i.e., at each $x \in \mathbb{R}$
- Standard every where diff. functions:**
- Polynomial function is every where differentiable
 - A constant function every where differentiable.
 - Exponential function a^x ($a > 0$, $a \neq 1$) is every where differentiable

Standard everywhere continuous functions:

- A constant function is every where continuous.
- Identity function is every where continuous.
- Modulus function is every where continuous.
- Exponential function
 $f(x) = a^x$ ($a > 0$, $a \neq 1$) is every where continuous.
- Polynomial function is every where continuous.

- In open interval (a,b) :
 $f(x)$ is cont. in (a,b)
iff it is cont. at every pt. between (a,b)
- In a closed interval $[a,b]$ if
- $f(x)$ is cont. in (a,b)
 - $\lim_{x \rightarrow a^+} f(x) = f(a)$
 - $\lim_{x \rightarrow b^-} f(x) = f(b)$

Every where continuous function:
 f is cont. in $(-\infty, \infty)$ i.e., on $\mathbb{R}$

Properties

Definition

Properties

Continuity

Differentiability

Discontinuity at a point

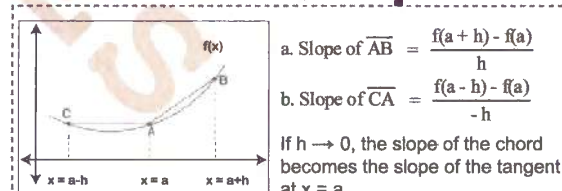
Differentiability at $x=a$ can be checked
if a unique tangent exists at $x=a$

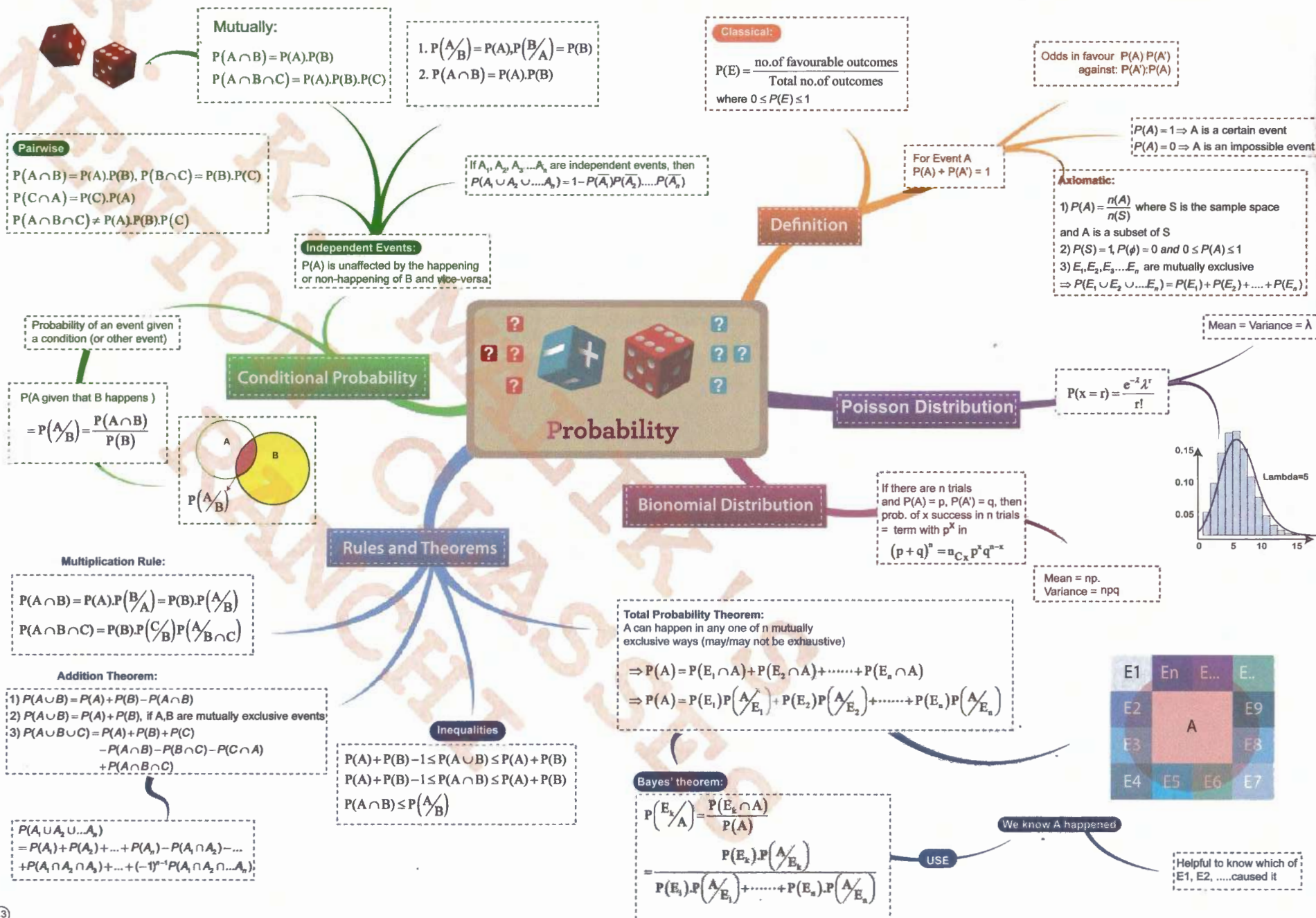
For a function to be diff at $x=a$,
it must be continuous
but not the converse

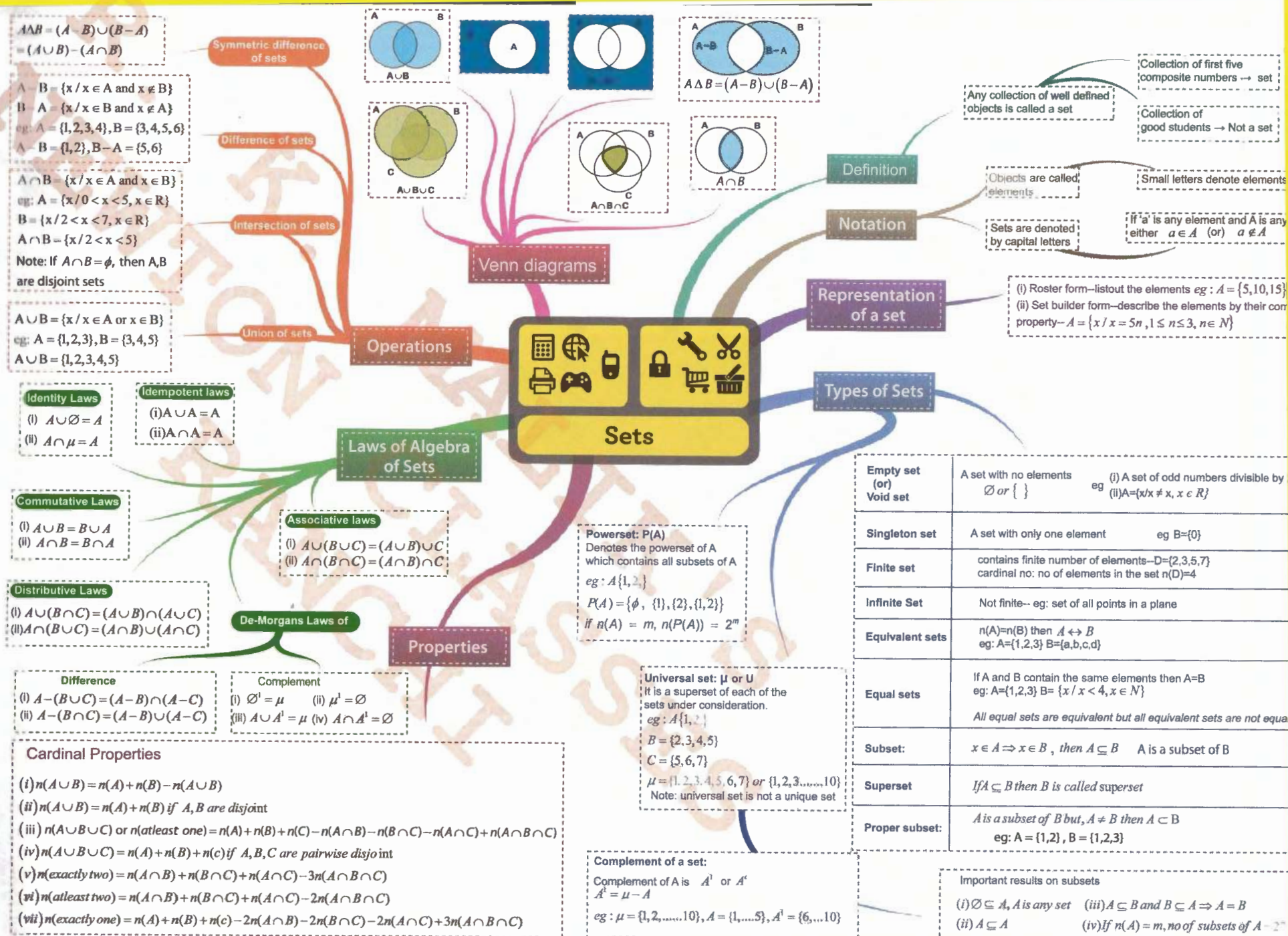
$f(c)$ exists
 $\Leftrightarrow f(c^+) = f(c^-)$
where $f(c^+) = \lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h}$
 $f(c^-) = \lim_{h \rightarrow 0} \frac{f(c-h) - f(c)}{-h}$

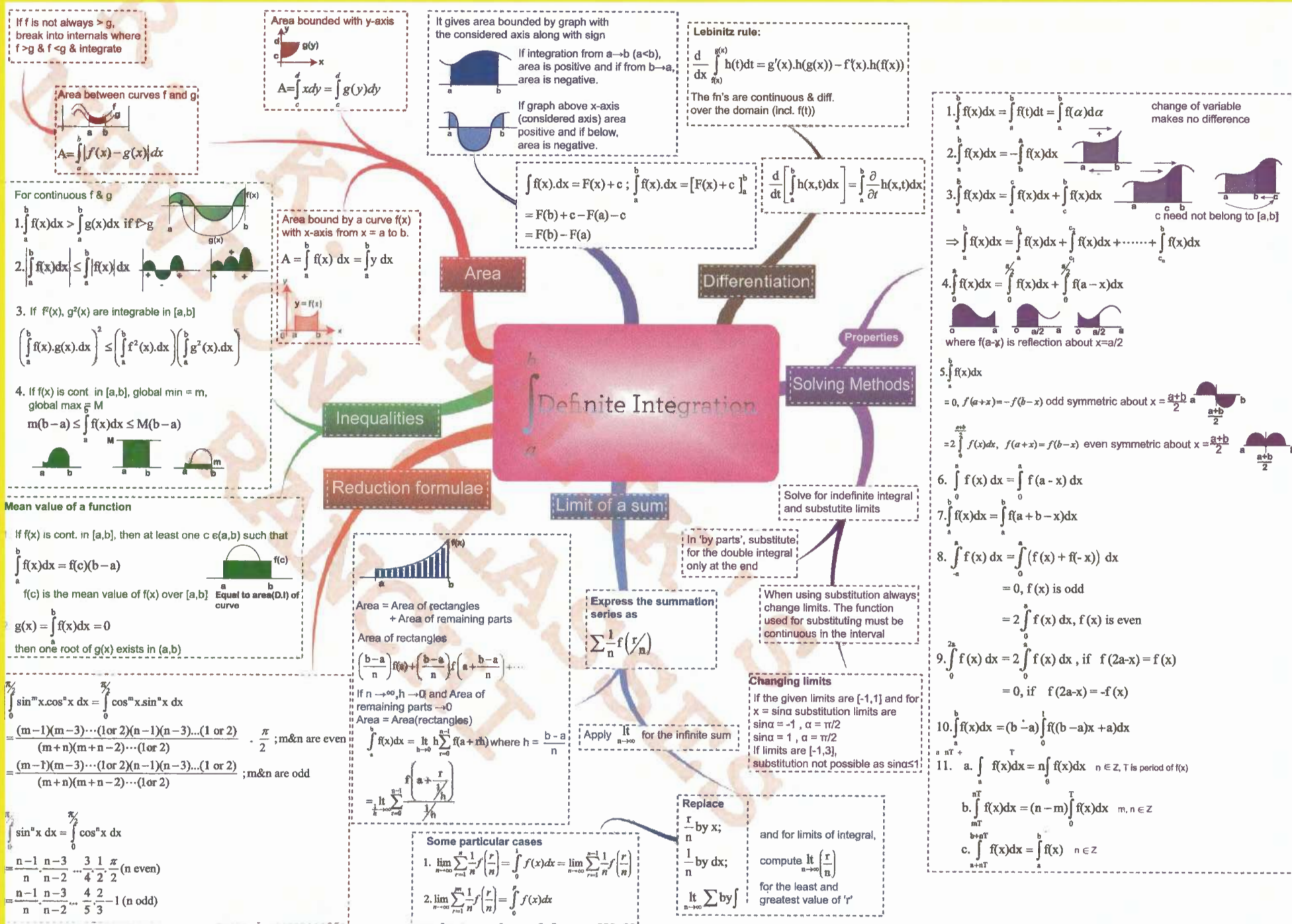
A function is diff. at $x=c$ if
 $\lim_{x \rightarrow c} \frac{f(x) - f(c)}{(x - c)}$ exists finitely

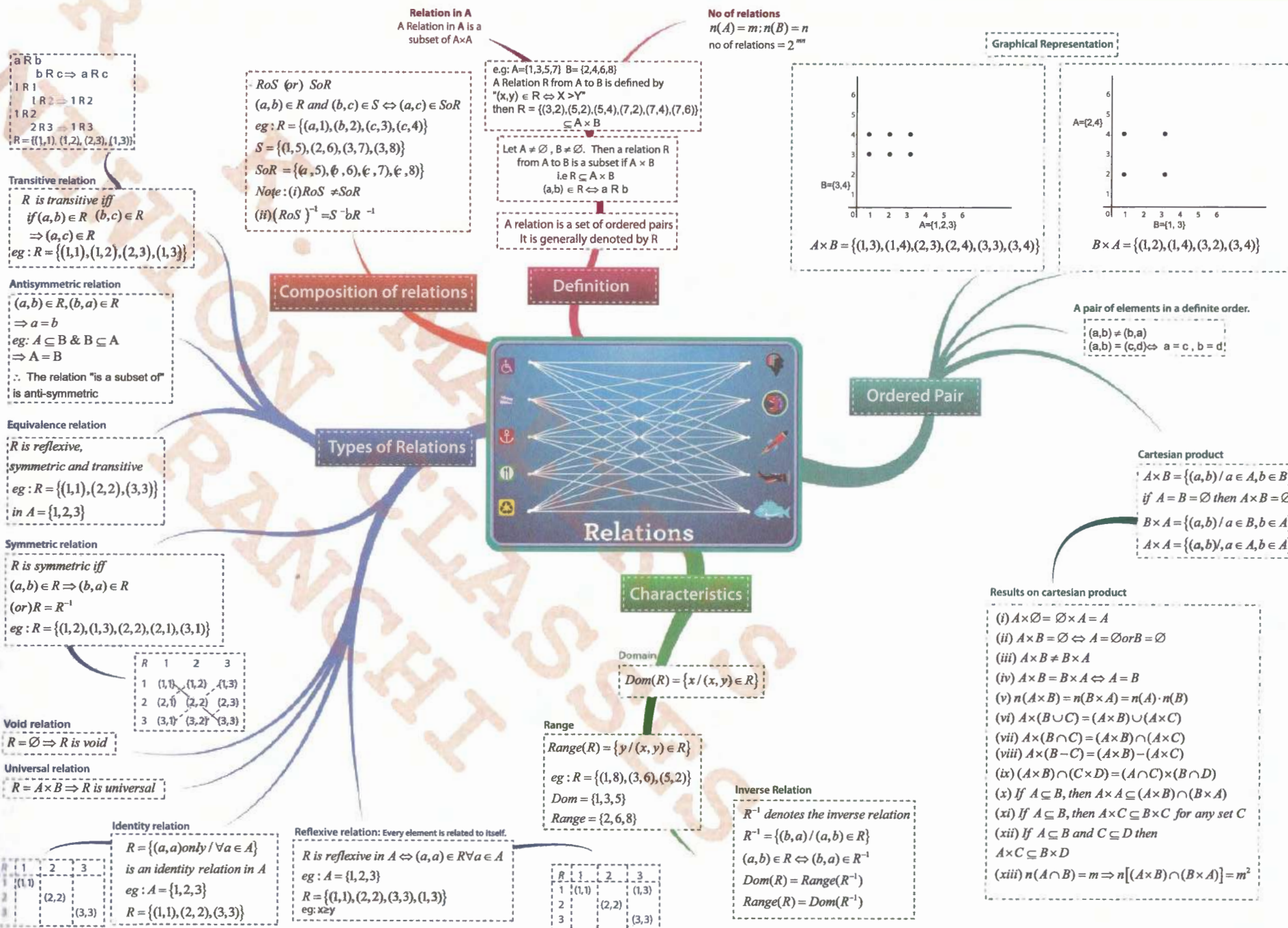
For a diff. function, its
derivative is given by
 $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

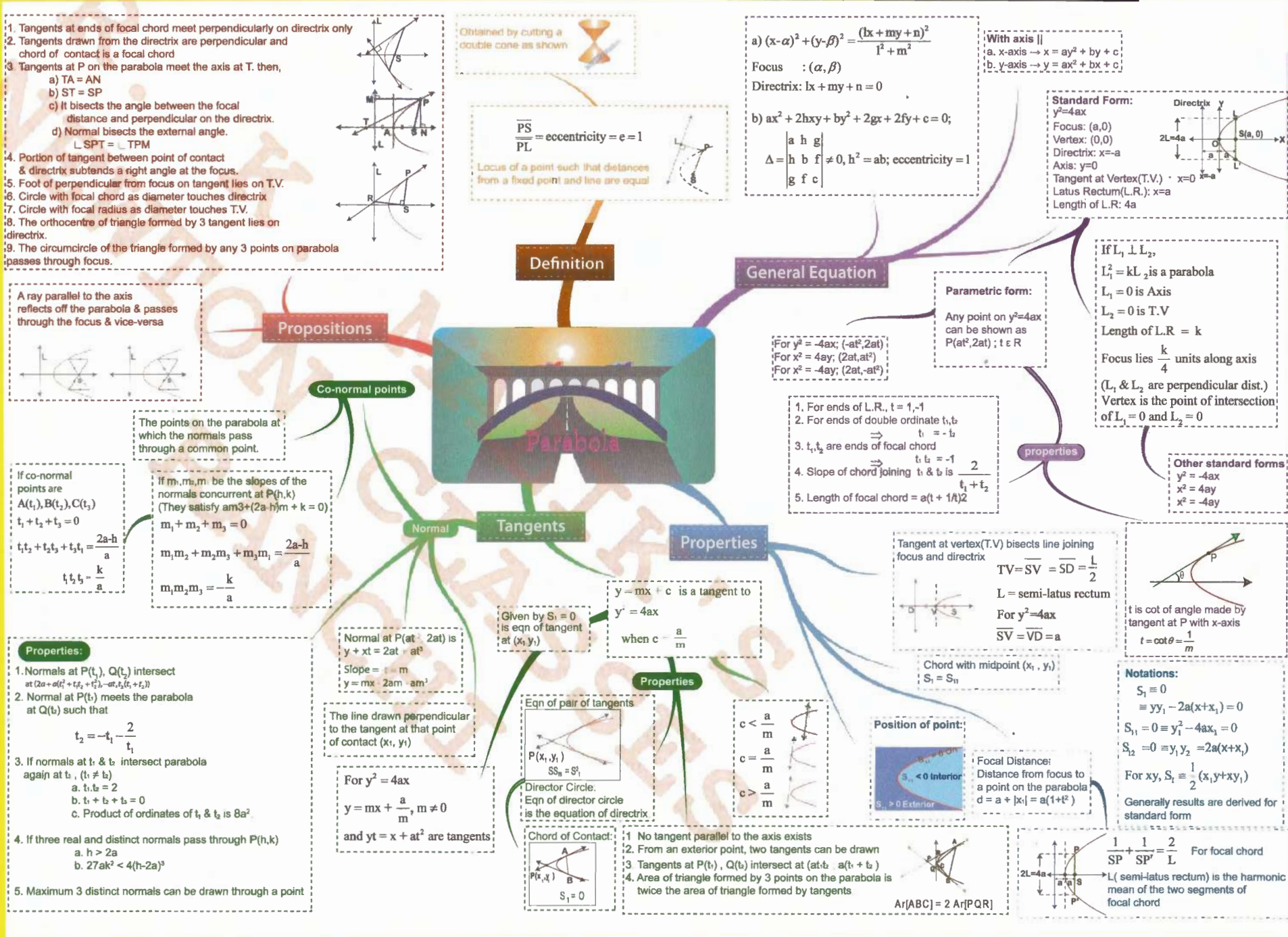












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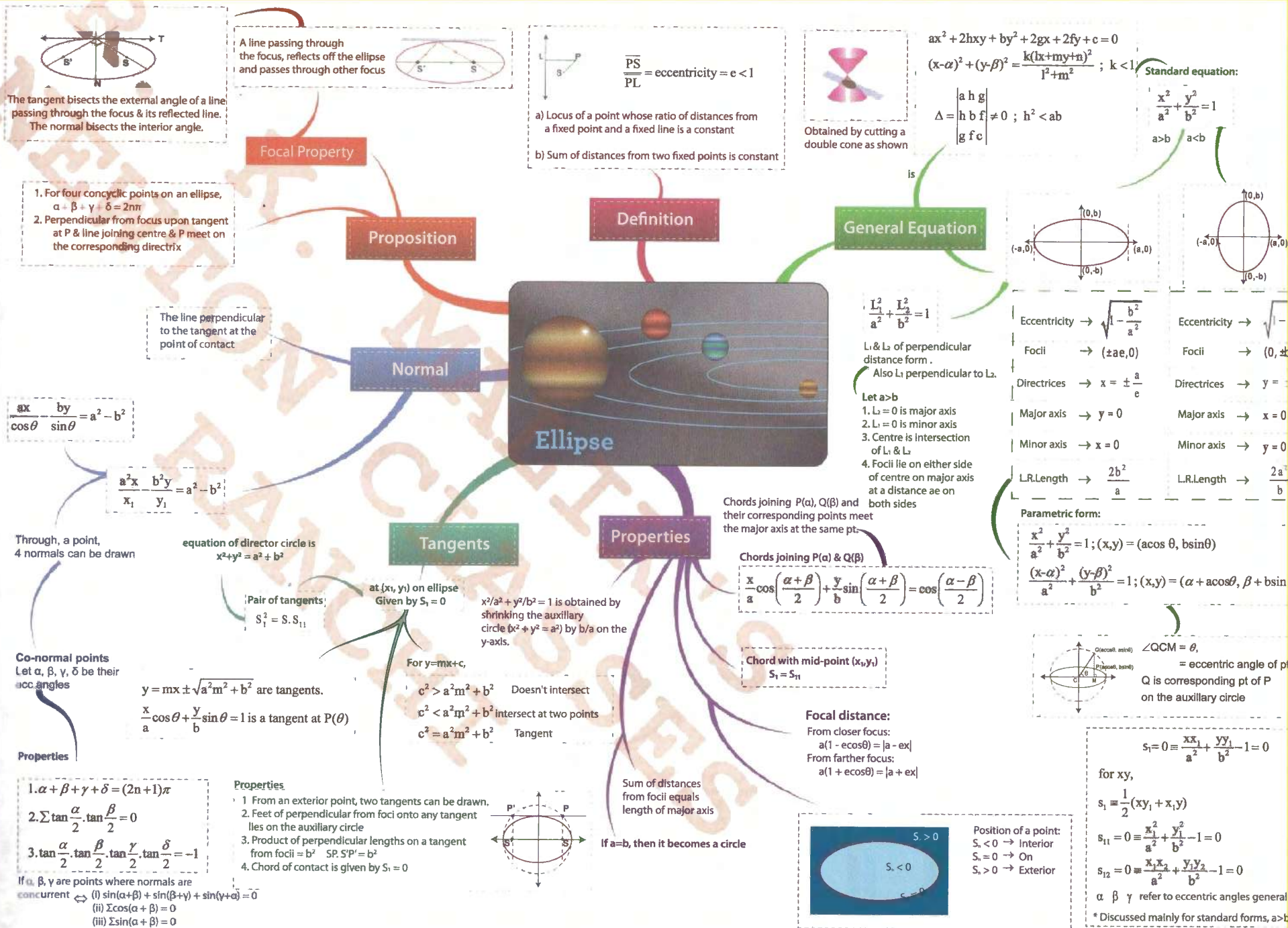
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Important Results

- $1+2+3+4+\dots+n = \sum n = \frac{n(n+1)}{2}$
- $1^2+2^2+3^2+4^2+\dots+n^2 = \sum n^2 = \frac{n(n+1)(2n+1)}{6}$
- $1^3+2^3+3^3+4^3+\dots+n^3 = \sum n^3 = \frac{n^2(n+1)^2}{4} = \left(\sum n\right)^2$
- $1^4+2^4+3^4+4^4+\dots+n^4 = \sum n^4 = \frac{n(n+1)(2n+1)(3n^2+3n-1)}{30}$
- $1+3+5+7+\dots+(2n-1) = \sum (2n-1) = n^2$
- $2+4+6+8+\dots+2n = \sum 2n = n(n+1)$
- $\sum n(n+1) = \frac{n(n+1)(n+2)}{3}$
- $\sum n(n+1)(n+2) = \frac{n(n+1)(n+2)(n+3)}{4}$
- The product of K consecutive integers is divisible by K!

- (i) Least Natural no. is 1
(ii) Set of Natural nos N is closed w.r.t addition of 1 to each of its elements.

Statements involving mathematical relations

Notation:
Use $p(n)$, $P(1)$, $P(2)$ to denote such statements

eg., $P(n)$: $n^3 + n$ is divisible by 3
 $P(3)$ is true
 $P(4)$ is false

Mathematical Statements

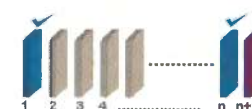
eg., $P(n)$: $n^2 > 100$
verify $P(r)$ is true
 $\Rightarrow P(r+1)$ is true

Sol: $P(r)$ is true
 $\Rightarrow r^2 > 100$
 $\Rightarrow r^2 + 2r + 1 > 100 + 2r + 1$
 $\Rightarrow (r+1)^2 > 100 + 2r + 1$
 $\Rightarrow (r+1)^2 > 100$ is true
(therefore $100 + 2r + 1 > 100$)
 $\Rightarrow P(r+1)$ is true

Basic Characteristics

First Principle of Induction

If (i) $P(n)$ is true for $n = 1$
ie., $P(1)$ is true
(ii) $P(m)$ is true
 $\Rightarrow P(m+1)$ is true
then $P(n)$ is true $\forall n \in \mathbb{N}$



Second Principle of Induction

$P(n)$: The no $(2+\sqrt{3})^n + (2-\sqrt{3})^n$ is an integer $\forall n \in \mathbb{N}$

$P(1)$: $2+\sqrt{3}+2-\sqrt{3} = 2 \in \mathbb{Z}$ is true
Let $P(m)$ be true $\forall m < k$

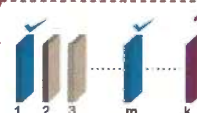
$P(k)$: $a^k + b^k$ where $a = 2+\sqrt{3}$; $b = 2-\sqrt{3}$
 $= (a^{k-1} + b^{k-1})(a+b) - ab(a^{k-2} + b^{k-2})$

$(a+b)$ and $ab \in \mathbb{Z}$ and also
 $a^{k-1} + b^{k-1}$ and $a^{k-2} + b^{k-2}$ are integers

$\therefore a^k + b^k \in \mathbb{Z}$

Thus $P(m)$ is true
 $\Rightarrow P(k)$ is true $\forall m < k$
 $\therefore P(n)$ is true $\forall n \in \mathbb{N}$

If (i) $P(n)$ is true for $n = 1$
(ii) $P(m)$ is true
 $\Rightarrow P(k)$ is true
 $\forall k \in \mathbb{N}$, $\forall m < k$
then $P(n)$ is true $\forall n \in \mathbb{N}$



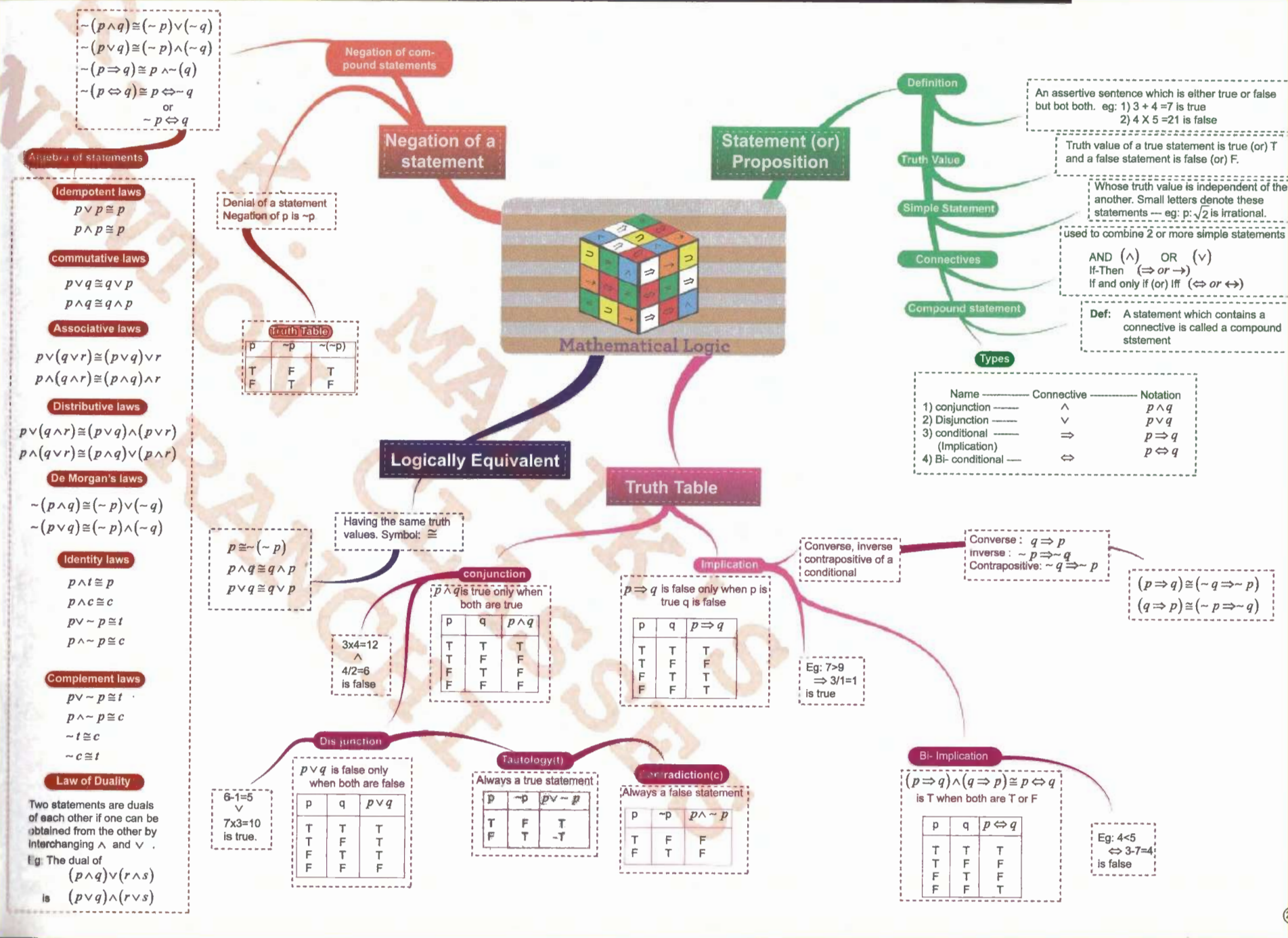
Third Principle of Induction

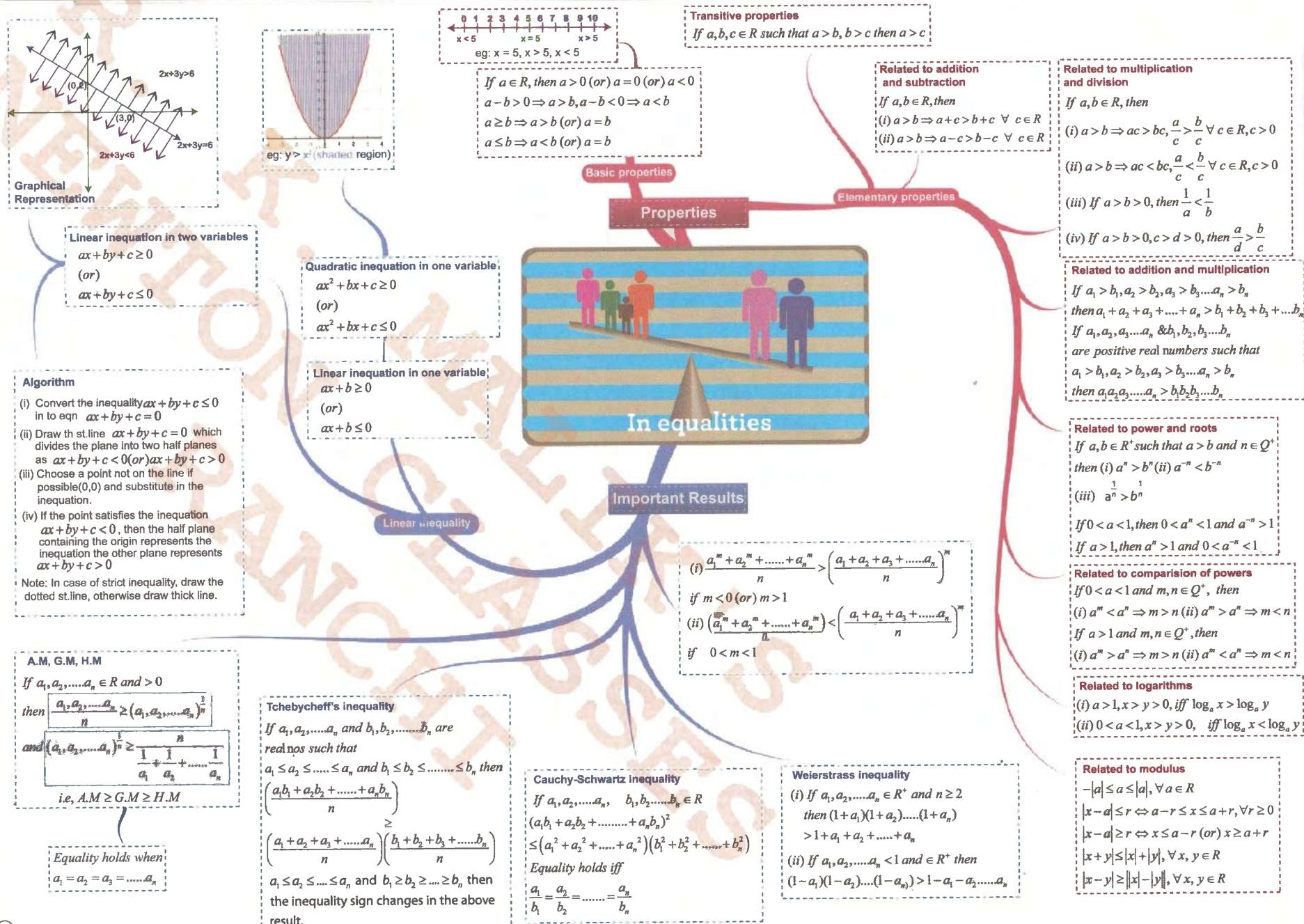
$P(n)$: $n^4 < 4^n$
 $P(1)$ is true
 $P(2)$ is false
 $P(3)$ is false
 $P(4)$ is false
 $P(5)$ is true

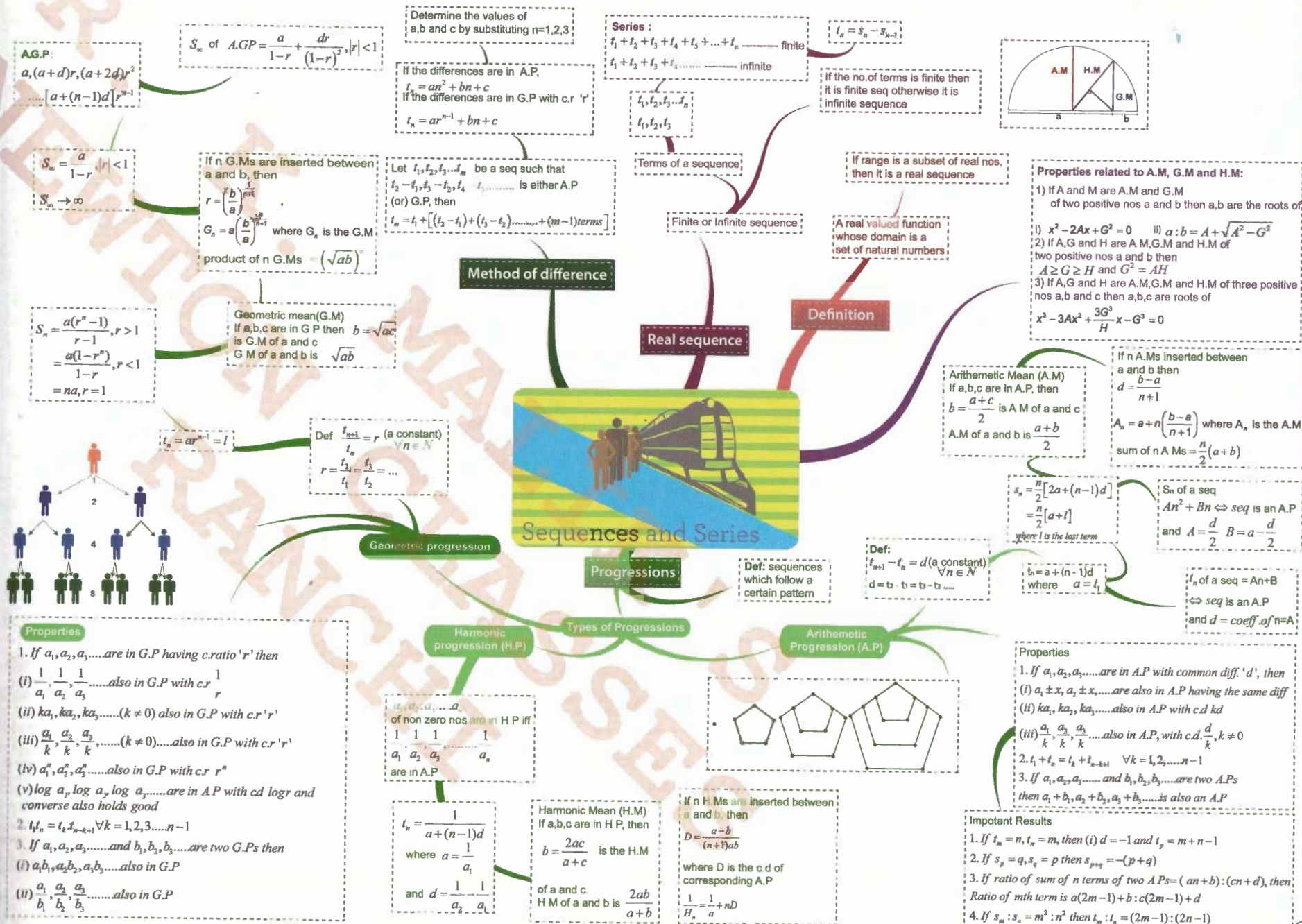
Let $P(k)$ be true
 $P(k)$ is true
 $\Rightarrow k^4 < 4^k$
 $\Rightarrow (k+1)^4 < 4^{k+1} \forall k \geq 5$
 $\therefore P(n)$ is true $\forall n \geq 5$

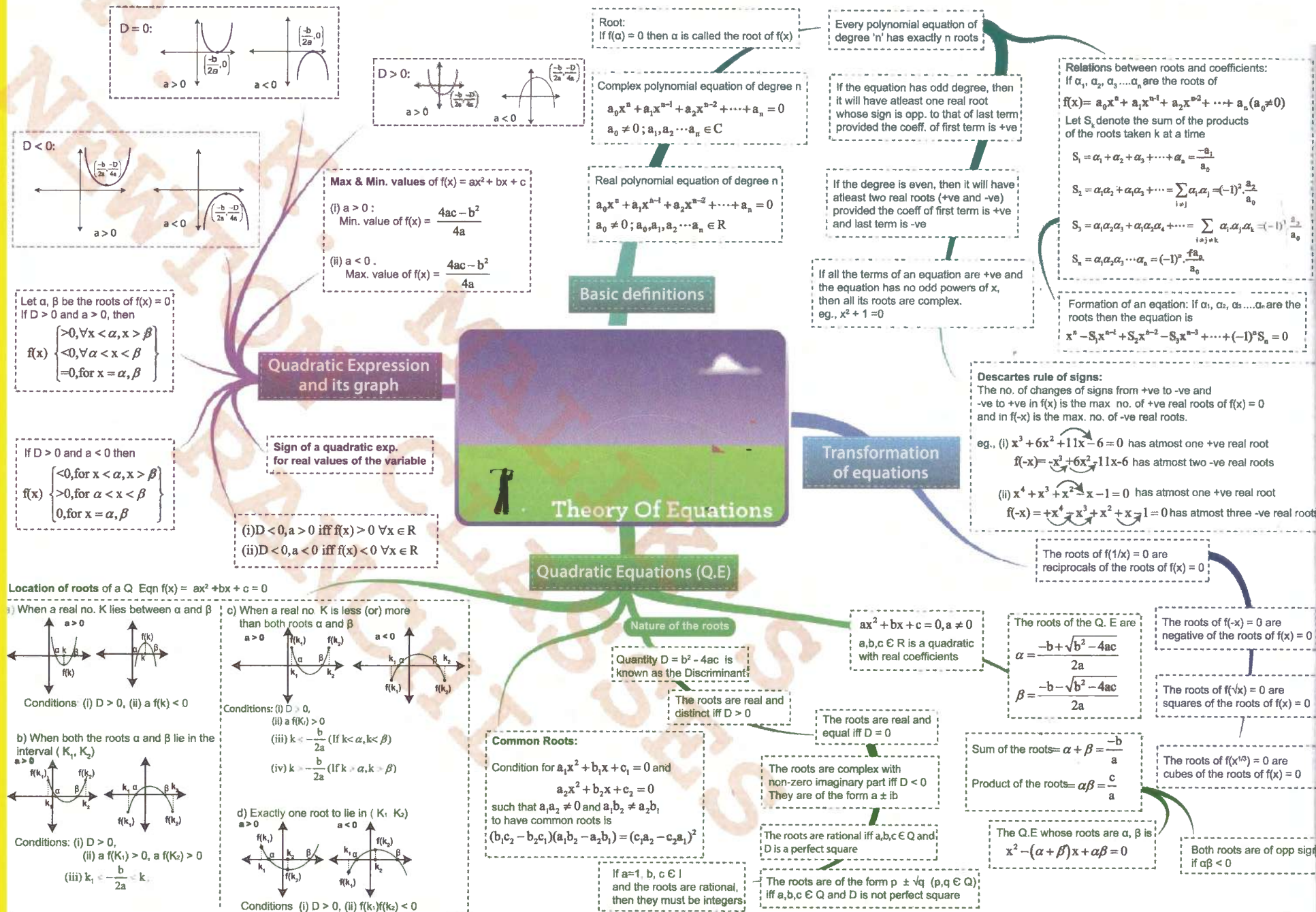
Let $P(n)$ be true statement
If
(i) $P(a)$ is true for some $a \in \mathbb{N}$
(ii) $P(k)$ is true $\Rightarrow P(k+1)$ is true $\forall a \leq k$
then $P(n)$ is true $\forall n \geq a$











Weighted Arithmetic mean

If $w_1, w_2, w_3, \dots, w_n$ are the weights assigned to $x_1, x_2, x_3, \dots, x_n$ then weighted A.M

$$\bar{X} = \frac{\sum w_i x_i}{\sum w_i}, \text{ where } i = 1, 2, \dots, n$$

eg:

x	w
1	1
2	2
3	3
4	4
.	.
.	.
.	.
n	n

Weighted A.M

$$= \frac{1^2 + 2^2 + \dots + n^2}{1 + 2 + \dots + n}$$

$$= \frac{2n+1}{3}$$

Raw data (or) ungrouped data (R.D)

$$\bar{X} = \frac{x_1 + x_2 + x_3 + \dots + x_n}{n}$$

eg: A.M of 1, 3, 5, 7, 9 is

$$\bar{X} = \frac{1+3+5+7+9}{5} = 5$$

Discrete frequency distribution (D.F.D)

x	f
x	f
x	f
x	f
x	f
x	f

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i}, \text{ where } i = 1, 2, \dots, n$$

Grouped Frequency Distribution (G.F.D)

$$\bar{X} = \frac{\sum f_i x_i}{N}$$

where x_i is the mid value of class interval and $N = \sum f_i$

Step Deviation Method

$$\bar{X} = A + \frac{\sum f_i d_i}{N} \times c \quad \text{where } A = \text{assumed mid value}$$

$$d_i = \frac{x_i - A}{c}$$

(OR) if class intervals of uniform length
C=length of C.I

$$\bar{X} = A + \frac{\sum f_i d_i}{N} \quad \text{Where } d_i = \frac{x_i - A}{c} \text{ if length of C.I is not uniform}$$

Properties of Arithmetic Mean

(i) Algebraic sum of deviations of a set of values from their A.M is zero $\rightarrow \sum f_i (x_i - \bar{X}) = 0$

(ii) Sum of squares of deviation about A.M is minimum $\rightarrow \sum f_i (x_i - A)^2$

is least when $A = \bar{X}$

(iii) Combined Arithmetic Mean

If $\bar{X}_1, \bar{X}_2, \dots, \bar{X}_K$ are the means of K series of sizes $n_1, n_2, \dots, n_K$ then

$$\bar{X} = \frac{n_1 \bar{X}_1 + n_2 \bar{X}_2 + \dots + n_K \bar{X}_K}{n_1 + n_2 + \dots + n_K}$$

For D.F.D

The value of x corresponding to cumulative frequency just greater than $N/2$

eg:

x	f	c.f
0	3	3
1	5	8
2	9	17
3	11	28
4	7	35
5	2	37

$$N = 37$$

$$\frac{N}{2} = 18.5$$

$$\text{Median} = 3$$

For R.D

If the data contains 'n' observations then

the median = $\left(\frac{n+1}{2}\right)^{\text{th}}$ observations, if 'n' is odd
= A.M. of $\left(\frac{n}{2}\right)^{\text{th}}$ and $\left(\frac{n}{2}+1\right)^{\text{th}}$ observation, if 'n' is even.

after arranging the data in ascending or descending order

eg: Median of 3, 5, 7, 9, 11, 13 is $\frac{7+9}{2} = 8$

For G.F.D

$$\text{Median} = l + \frac{\frac{N}{2} - F}{f} \times c$$

Median class is the class corresponding to c.f just greater than $N/2$

where l = lower boundary of median class
F = C.F of Median class
f = frequency of median class
c = length of median class

For G.F.D

$$\text{Mode} = l + \frac{f - f_1}{2f - f_1 - f_2} \times c$$

Class corresponding to highest frequency is the modal class

l = lower boundary of modal class

f = frequency of modal class

f_1 = frequency of class preceeding modal class

f_2 = frequency of class succeeding modal class

c = length of modal class

For R.D and For D.F.D

The value of the variable for which the frequency is maximum.

eg: Mode of 3, 5, 2, 3, 4, 5, 3, 5, 6, 5, 7 is 5

eg:

x	f
0	3
1	4
2	7
3	5
4	1

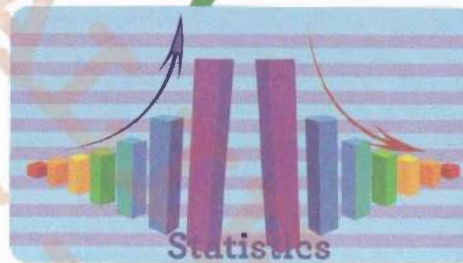
mode is 2

Measures of Central Tendency

Arithmetic Mean (A.M)

Median

Mode



Measure of Dispersion

Mean Deviation

For R.D $M.D. = \frac{\sum |x_i - A|}{n}$

For G.F.D $M.D. = \frac{1}{N} \sum f_i |x_i - A|$

where x is the mid value of the class and A is A.M or median or mode.

Standard Results

(1) If the S.D of a variate X is σ , then S.D of $aX + b$ is $|a|\sigma$

(2) If the S.D of a variate X is σ , then S.D of $\frac{aX+b}{c}$ is $\frac{|a|}{|c|}\sigma$

(3) S.D of first n natural no.s is $\frac{n^2-1}{12}$

(4) M.D from the mean of the A.P $a, a+d, a+2d, \dots, a+2nd$ is $\frac{n(n+1)}{2n+1}d$

(5) S.D from the mean of the A.P $a, a+d, a+2d, \dots, a+2nd$ is $\sqrt{\frac{n(n+1)}{3}}d$

Standard deviation

Square root of the average of squares of deviation taken from A.M is the S.D and is denoted by σ

Property of S.D

i. S.D is independent of change of origin

eg: If S.D of $x_1, x_2, x_3, \dots, x_n$ is σ

then S.D of $x_1+k, x_2+k, x_3+k, \dots, x_n+k$ is also σ

ii. S.D is not independent of change of scale

eg: If S.D of $x_1, x_2, x_3, \dots, x_n$ is σ then

S.D of $Kx_1, Kx_2, Kx_3, \dots, Kx_n$ is $|K|\sigma$

iii. the square of standard deviation is called variance

$$\text{variance} = \sigma^2$$

For R.D

$$S.D (\sigma) = \sqrt{\frac{\sum (x_i - \bar{X})^2}{n}}$$

For G.F.D

$$S.D (\sigma) = \sqrt{\frac{1}{N} \sum f_i (x_i - \bar{X})^2}$$

where x_i is the mid value of the class

$$= \sqrt{\frac{\sum f_i x_i^2}{N} - \left(\frac{\sum f_i x_i}{N}\right)^2}$$

$$= \sqrt{\frac{\sum f_i d_i^2}{N} - \left(\frac{\sum f_i d_i}{N}\right)^2}$$

Variance of the combined series

If n_1, n_2 are the sizes; $\bar{X}_1, \bar{X}_2$ are the means and σ_1, σ_2 are the standard deviations of two series then variance of combined series is

$$\sigma^2 = \frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{n_1 + n_2}$$

$$\bar{X} = \frac{n_1 \bar{X}_1 + n_2 \bar{X}_2}{n_1 + n_2} \text{ and}$$

$$\text{where } d_1 = \bar{X} - \bar{X}_1, d_2 = \bar{X}_2 - \bar{X}$$

(i) $\sum_{n=0}^{\infty} \frac{1}{n!} = e = \sum_{n=0}^{\infty} \frac{1}{(n-K)!} = e = \sum_{n=0}^{\infty} \frac{1}{(n-1)!}$

(ii) $\sum_{n=1}^{\infty} \frac{1}{n!} = e - 1$

(iii) $\sum_{n=2}^{\infty} \frac{1}{n!} = e - 2$

(iv) $\sum_{n=0}^{\infty} \frac{1}{(2n)!} = \sum_{n=1}^{\infty} \frac{1}{(2n-2)!} = \frac{e+e^{-1}}{2}$

(v) $\sum_{n=0}^{\infty} \frac{1}{(2n+1)!} = \sum_{n=1}^{\infty} \frac{1}{(2n-1)!} = \frac{e-e^{-1}}{2}$

(vi) $\sum_{n=0}^{\infty} \frac{1}{(n+1)!} = e - 1$

(vii) $\sum_{n=1}^{\infty} \frac{1}{(n+1)!} = e - 2$

(viii) $\sum_{n=0}^{\infty} \frac{1}{(n+2)!} = e - 2$

(ix) $\sum_{n=1}^{\infty} \frac{1}{(n+2)!} = e - \frac{5}{2}$

(x) $\sum_{n=1}^{\infty} \frac{1}{(n-1)!} = \sum_{n=1}^{\infty} \frac{1}{(n-2)!} = e$

(xi) $\sum_{n=1}^{\infty} \frac{n}{n!} = \sum_{n=1}^{\infty} \frac{1}{(n-1)!} = e$

(xii) $\sum_{n=1}^{\infty} \frac{n^2}{n!} = 2e$

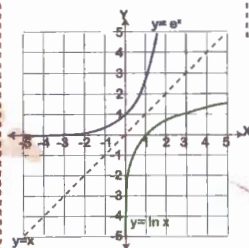
(xiii) $\sum_{n=1}^{\infty} \frac{n^3}{n!} = 5e$

(xiv) $\sum_{n=1}^{\infty} \frac{n^4}{n!} = 15e$

(xv) $\sum_{n=0}^{\infty} \frac{x^n}{n!} = \sum_{n=1}^{\infty} \frac{x^{n-1}}{(n-1)!} = \sum_{n=r}^{\infty} \frac{x^{n-r}}{(n-r)!} = e^x$

(xvi) $\sum_{n=1}^{\infty} \frac{x^n}{(n-1)!} = xe^x$

(xvii) $\sum_{n=r}^{\infty} \frac{x^n}{(n-r)!} = x^r e^x$



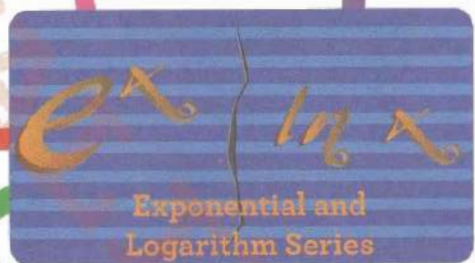
Graphical Representation

Important Results

Logarithm Series

General term and coefficient

Exponential Series



T_{n+1} in $e^x = \frac{x^n}{n!}$
coeff. of $x^n = \frac{1}{n!}$

T_{n+1} in $e^{-x} = \frac{(-1)^n x^n}{n!}$
coeff. of $x^n = \frac{(-1)^n}{n!}$

T_{n+1} in $e^{ax} = \frac{(ax)^n}{n!}$
coeff. of $x^n = \frac{a^n}{n!}$

The limiting value of the expression
 $\left(1 + \frac{1}{n}\right)^n$ as $n \rightarrow \infty$ is e

For any $x \in \mathbb{R}$, we have
 $e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \dots + \frac{x^n}{n!} + \dots \infty$

$e = 1 + \frac{1}{1!} + \frac{1}{2!} + \dots + \frac{1}{n!} + \dots \infty$

$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} \dots (i)$

$e^{-x} = 1 - \frac{x}{1!} + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots + \frac{(-1)^n x^n}{n!} + \dots \infty$

$e^{-x} = \sum_{n=0}^{\infty} \frac{(-1)^n x^n}{n!}$

$\frac{e^x + e^{-x}}{2} = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} + \dots \infty$

$\frac{e^x + e^{-x}}{2} = \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}$

$\frac{e^x - e^{-x}}{2} = \frac{x}{1!} + \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!} + \dots \infty$

$\frac{e^x - e^{-x}}{2} = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}$

$e^{-1} = 1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \dots \infty$

$e^{-1} = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!}$

$\frac{e + e^{-1}}{2} = 1 + \frac{1}{2!} + \frac{1}{4!} + \frac{1}{6!} + \dots \infty$

$\frac{e + e^{-1}}{2} = \sum_{n=0}^{\infty} \frac{1}{(2n)!}$

$\frac{e - e^{-1}}{2} = 1 + \frac{1}{3!} + \frac{1}{5!} + \frac{1}{7!} + \dots \infty$

$\frac{e - e^{-1}}{2} = \sum_{n=0}^{\infty} \frac{1}{(2n+1)!}$

$e^{ax} = 1 + \frac{ax}{1!} + \frac{a^2 x^2}{2!} + \frac{a^3 x^3}{3!} + \dots + \frac{(ax)^n}{n!} + \dots \infty$

$e^{ax} = \sum_{n=0}^{\infty} \frac{(ax)^n}{n!}$

Let $a > 0, \forall x \in \mathbb{R}$

$a^x = 1 + x \log_e a + \frac{x^2}{2!} (\log_e a)^2 + \frac{x^3}{3!} (\log_e a)^3 + \dots + \frac{x^n}{n!} (\log_e a)^n + \dots \infty$
 $= \sum_{n=0}^{\infty} \frac{(x \log_e a)^n}{n!}$

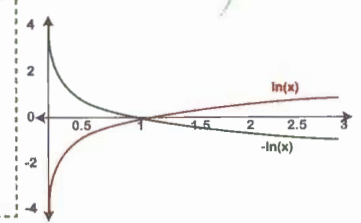
$T_{n+1} = \frac{x^n (\log_e a)^n}{n!}$
coeff. of $x^n = \frac{(\log_e a)^n}{n!}$

If $|x| < 1$, then
 $\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots \infty$
 $= \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n}$
 $\log 2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \dots \infty$
 $\log(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \dots \infty$
 $= -\sum_{n=1}^{\infty} \frac{x^n}{n}$
 $-\log(1-x) = x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \dots \infty$
 $= \sum_{n=1}^{\infty} \frac{x^n}{n}$
 $\log\left(\frac{1+x}{1-x}\right) = 2\left(x + \frac{x^3}{3} + \frac{x^5}{5} + \dots \infty\right)$
 $\log(1-x^2) = -2\left(\frac{x^2}{2} + \frac{x^4}{4} + \frac{x^6}{6} + \dots \infty\right)$

Important Application

coeff. of x^n in $\log_e(1+x+x^2)$
 $\log(1+x+x^2) = \log[(1-\omega x)(1-\omega^2 x)]$
where ω, ω^2 are cube roots of unity
 $= \log(1-\omega x) + \log(1-\omega^2 x)$

Coeff. of $x^n = \begin{cases} -\frac{1}{n} (\omega^n + \omega^{2n}) & \text{if } n \neq 3k \\ \frac{1}{n} & \text{if } n = 3k \\ -\frac{2}{n} & \text{if } n = 3k \end{cases}$



Function(y)	Domain	Range	Principal value branch
$\sin^{-1}x$	$[-1, 1]$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$	$y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
$\cos^{-1}x$	$[-1, 1]$	$[0, \pi]$	$y \in [0, \pi]$
$\tan^{-1}x$	$\mathbb{R}$	$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$	$y \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
$\cot^{-1}x$	$\mathbb{R}$	$(0, \pi)$	$y \in (0, \pi)$
$\operatorname{cosec}^{-1}x$	$\mathbb{R} - (-1, 1)$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - \{0\}$	$y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]; y \neq 0$
$\sec^{-1}x$	$\mathbb{R} - (-1, 1)$	$[0, \pi] - \left\{\frac{\pi}{2}\right\}$	$y \in [0, \pi]; y \neq \frac{\pi}{2}$

(i) $\sin^{-1}(\sin\theta) = \theta; \forall \theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$	(iv) $\operatorname{cosec}^{-1}(\operatorname{cosec}\theta) = \theta; \forall \theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - \{0\}$
(ii) $\cos^{-1}(\cos\theta) = \theta; \forall \theta \in [0, \pi]$	(v) $\sec^{-1}(\sec\theta) = \theta; \forall \theta \in [0, \pi] - \left\{\frac{\pi}{2}\right\}$
(iii) $\tan^{-1}(\tan\theta) = \theta; \forall \theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$	(vi) $\cot^{-1}(\cot\theta) = \theta; \forall \theta \in (0, \pi)$

- (i) $\sin(\sin^{-1}x) = x; \forall x \in [-1, 1]$
 (ii) $\cos(\cos^{-1}x) = x; \forall x \in [-1, 1]$
 (iii) $\tan(\tan^{-1}x) = x; \forall x \in \mathbb{R}$
 (iv) $\cot(\cot^{-1}x) = x; \forall x \in \mathbb{R}$
 (v) $\operatorname{cosec}(\operatorname{cosec}^{-1}x) = x; \forall x \in \mathbb{R} - (-1, 1)$
 (vi) $\sec(\sec^{-1}x) = x; \forall x \in \mathbb{R} - (-1, 1)$

- (i) $\sin^{-1}(-x) = -\sin^{-1}x; \forall x \in [-1, 1]$
 (ii) $\cos^{-1}(-x) = \pi - \cos^{-1}x; \forall x \in [-1, 1]$
 (iii) $\tan^{-1}(-x) = -\tan^{-1}x; \forall x \in \mathbb{R}$
 (iv) $\operatorname{cosec}^{-1}(-x) = -\operatorname{cosec}^{-1}x; \forall x \in \mathbb{R} - (-1, 1)$
 (v) $\sec^{-1}(-x) = \pi - \sec^{-1}x; \forall x \in \mathbb{R} - (-1, 1)$
 (vi) $\cot^{-1}(-x) = \pi - \cot^{-1}x; \forall x \in \mathbb{R}$

Property - XII

- (i) if $|x| \leq 1$, then

$$2\tan^{-1}x = \sin^{-1}\frac{2x}{1+x^2}$$

$$= \cos^{-1}\frac{1-x^2}{1+x^2}$$

$$= \tan^{-1}\frac{2x}{1-x^2}$$
 (ii) if $|x| > 1$, then
 change x into $\frac{1}{x}$ in the above

Property - XI

- (i) $3\sin^{-1}x = \sin^{-1}(3x - 4x^3)$
 (ii) $3\cos^{-1}x = \cos^{-1}(4x^3 - 3x)$
 (iii) $3\tan^{-1}x = \tan^{-1}\left(\frac{3x - x^3}{1 - 3x^2}\right)$

Property - X

- (i) $2\sin^{-1}x = \sin^{-1}2x\sqrt{1-x^2}$
 (ii) $2\cos^{-1}x = \cos^{-1}(2x^2 - 1)$
 (iii) $2\tan^{-1}x = \tan^{-1}\frac{2x}{1-x^2}$

Property - I

Property - II

Property - III

Basic Definition



Inverse Trigonometric functions

Properties of inverse functions

Property - IV

- (i) $\sin^{-1}\frac{1}{x} = \operatorname{cosec}^{-1}x; \forall x \in \mathbb{R} - (-1, 1)$
 (ii) $\cos^{-1}\frac{1}{x} = \sec^{-1}x; \forall x \in \mathbb{R} - (-1, 1)$
 (iii) $\tan^{-1}\frac{1}{x} = \cot^{-1}x; x > 0$
 $= -\pi + \cot^{-1}x; x < 0$

Property - V

- (i) $\sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}; \forall x \in [-1, 1]$
 (ii) $\tan^{-1}x + \cot^{-1}x = \frac{\pi}{2}; \forall x \in \mathbb{R}$
 (iii) $\operatorname{cosec}^{-1}x + \sec^{-1}x = \frac{\pi}{2}; \forall x \in \mathbb{R} - (-1, 1)$

Property - VI

- (i) $\tan^{-1}x + \tan^{-1}y = \tan^{-1}\left(\frac{x+y}{1-xy}\right); \text{if } xy < 1$
 $= \pi + \tan^{-1}\left(\frac{x+y}{1-xy}\right); \text{if } x, y > 0; xy > 1$
 $= -\pi + \tan^{-1}\left(\frac{x+y}{1-xy}\right); \text{if } x, y < 0; xy > 1$

Property - VII

- (ii) $\tan^{-1}x - \tan^{-1}y = \tan^{-1}\left(\frac{x-y}{1+xy}\right)$

Property - VIII

- (i) $\cos^{-1}x + \cos^{-1}y = \cos^{-1}\{xy - \sqrt{1-x^2}\sqrt{1-y^2}\}$
 (ii) $\cos^{-1}x - \cos^{-1}y = \cos^{-1}\{xy + \sqrt{1-x^2}\sqrt{1-y^2}\}$

Property - IX

- (i) $\sin^{-1}x = \cos^{-1}\sqrt{1-x^2} = \tan^{-1}\frac{x}{\sqrt{1-x^2}}$
 $= \cot^{-1}\frac{\sqrt{1-x^2}}{x} = \sec^{-1}\frac{1}{\sqrt{1-x^2}} = \operatorname{cosec}^{-1}\frac{1}{x}$

- (ii) $\cos^{-1}x = \sin^{-1}\sqrt{1-x^2} = \tan^{-1}\frac{\sqrt{1-x^2}}{x}$
 $= \cot^{-1}\frac{x}{\sqrt{1-x^2}} = \sec^{-1}\frac{1}{x} = \operatorname{cosec}^{-1}\frac{1}{\sqrt{1-x^2}}$

- (iii) $\tan^{-1}x = \sin^{-1}\frac{x}{\sqrt{1+x^2}} = \cos^{-1}\frac{1}{\sqrt{1+x^2}}$
 $= \cot^{-1}\frac{1}{x} = \sec^{-1}\sqrt{1+x^2} = \operatorname{cosec}^{-1}\frac{\sqrt{1+x^2}}{x}$

I. Solution of a right angled triangle (suppose $\angle C = 90^\circ$)

Case 1: When two sides are given

Given	Required
(i) a, b	$\tan A = \frac{a}{b}, \angle B = 90^\circ - A, c = \frac{a}{\sin A}$
(ii) a, c	$\sin A = \frac{a}{c}, b = c \cos A, \angle B = 90^\circ - A$

Case 2: When a side and acute angle are given

Given	Required
(i) a, A	$B = 90^\circ - A, b = a \cot A, c = \frac{a}{\sin A}$
(ii) c, A	$B = 90^\circ - A, a = c \sin A, b = c \cos A$

II. Solution of a triangle in general

Case 1: When three sides a, b, c are given

$$\Delta = \sqrt{s(s-a)(s-b)(s-c)}$$

$$\sin A = \frac{2\Delta}{bc}, \sin B = \frac{2\Delta}{ac}, \sin C = \frac{2\Delta}{ab}$$

Case 2: When two sides a, b and included angle C are given

$$c = \frac{a \sin C}{\sin A}, \tan \frac{A-B}{2} = \frac{a-b}{a+b} \cot \frac{C}{2}$$

$$\tan \frac{A+B}{2} = \cot \frac{C}{2}$$

Case 3: When one side a and two angles A and B are given

$$C = 180^\circ - (A+B), b = \frac{a \sin B}{\sin A}, c = \frac{a \sin C}{\sin A}$$

Case 4: When two sides a, b and the angle opposite to third C are given

$$\sin B = \frac{b \sin A}{a}, \angle C = 180^\circ - (A+B), c = \frac{a \sin C}{\sin A}$$

- A is acute, $b \sin A > a \Rightarrow \sin B > 1$, which is impossible
- A is acute, $b \sin A = a \Rightarrow$ triangle with $\angle B = 90^\circ$
- A is acute, $b \sin A < a \Rightarrow$ two values of $B - B_1$ & B_2 such that $B_1 + B_2 = 180^\circ$

Length of angle bisectors

- Length of angle bisector of $\angle A = \frac{2bc}{b+c} \cos \frac{A}{2}$
- Length of angle bisector of $\angle B = \frac{2ca}{c+a} \cos \frac{B}{2}$
- Length of angle bisector of $\angle C = \frac{2ab}{a+b} \cos \frac{C}{2}$

In ΔABC

- $\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} = \frac{1}{r}$
- $r_1 + r_2 + r_3 - r = 4R$
- $r_1^2 + r_2^2 + r_3^2 + r^2 = 16R^2 - (a^2 + b^2 + c^2)$
- $\cos A + \cos B + \cos C = 1 + \frac{r}{R}$
- $\sin A + \sin B + \sin C = \frac{s}{R}$
- $\Delta^2 = r r_1 r_2 r_3$
- $\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} + \frac{1}{r} = \frac{a^2 + b^2 + c^2}{\Delta^2}$

Length of medians:

- On side $BC = \sqrt{\frac{1}{4}(2b^2 + 2c^2 - a^2)}$
- On side $AC = \sqrt{\frac{1}{4}(2c^2 + 2a^2 - b^2)}$
- On side $AB = \sqrt{\frac{1}{4}(2a^2 + 2b^2 - c^2)}$

m-n rule:

- $(m+n) \cot \theta = m \cot A + n \cot B$
- $(m+n) \cot \theta = n \cot B - m \cot C$



Let a be the side of the regular polygon and n be the no. of sides

$$R = \frac{a}{2} \operatorname{cosec} \frac{\pi}{n}$$

$$r = \frac{a}{2} \cot \frac{\pi}{n}$$

$$\left\{ \begin{array}{l} \frac{n}{4} a^2 \cot \frac{\pi}{n} \\ n r^2 \tan \frac{\pi}{n} \\ \frac{n R^2}{2} \sin \frac{2\pi}{n} \end{array} \right.$$

Area of regular Polygon

Regular polygon and radius of inscribed and circumscribed circles about it

$$1. \text{ Sine rule: } \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = K \text{ (K is a constant)}$$

$$2. \text{ cosine rule: } a^2 = b^2 + c^2 - 2bc \cos A \Rightarrow \cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

Napier's Analogy (tangent rule):

$$\tan \left(\frac{B-C}{2} \right) = \frac{b-c}{b+c} \cot \frac{A}{2} \quad \tan \left(\frac{C-A}{2} \right) = \frac{c-a}{c+a} \cot \frac{B}{2}$$

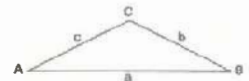
$$\tan \left(\frac{A-B}{2} \right) = \frac{a-b}{a+b} \cot \frac{C}{2}$$

$$\frac{\sin \left(\frac{B-C}{2} \right)}{\cos \frac{A}{2}} = \frac{b-c}{a}$$

$$\frac{\cos \left(\frac{B-C}{2} \right)}{\sin \frac{A}{2}} = \frac{b+c}{a}$$

3. Projection rule:

$$c = a \cos B + b \cos A$$



Important Points

Solutions of triangles

Properties & Solutions of Triangles

Rules

Half angle formulae

$$\begin{aligned} \sin \frac{A}{2} &= \sqrt{\frac{(s-b)(s-c)}{bc}} \\ \cos \frac{A}{2} &= \sqrt{\frac{s(s-a)}{bc}} \\ \tan \frac{A}{2} &= \sqrt{\frac{(s-b)(s-c)}{s(s-a)}} \\ &= \frac{\Delta}{s(s-a)} \\ &= \frac{(s-b)(s-c)}{\Delta} \end{aligned}$$

Area (Δ)

$$\begin{aligned} \Delta &= \frac{1}{2} bc \sin A \\ &= \frac{1}{2} ca \sin B \\ &= \frac{1}{2} ab \sin C \\ &= \sqrt{s(s-a)(s-b)(s-c)} \end{aligned}$$



Circumcentre and circumcircle

Incentre and incircle

Excentre and excircle

Cyclic Quadilateral

Pedal Triangle

Orthocentre

The circumcentre(S), ortho centre(O), and Centroid(G) are collinear

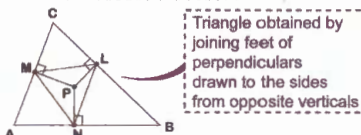
$$OG : GS = 2 : 1$$

$$OS = R \sqrt{1 - 8 \cos A \cos B \cos C}$$

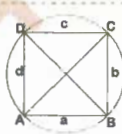
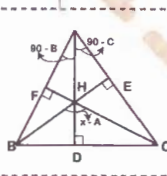
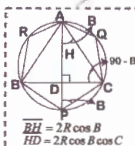
$$OI = \sqrt{2r^2 - 4R^2 \cos A \cos B \cos C}$$

$$SI = \sqrt{R^2 - 2Rr}$$

$$OG = \sqrt{R^2 - \frac{a^2 + b^2 + c^2}{9}}$$



Triangle obtained by joining feet of perpendiculars drawn to the sides from opposite vertices



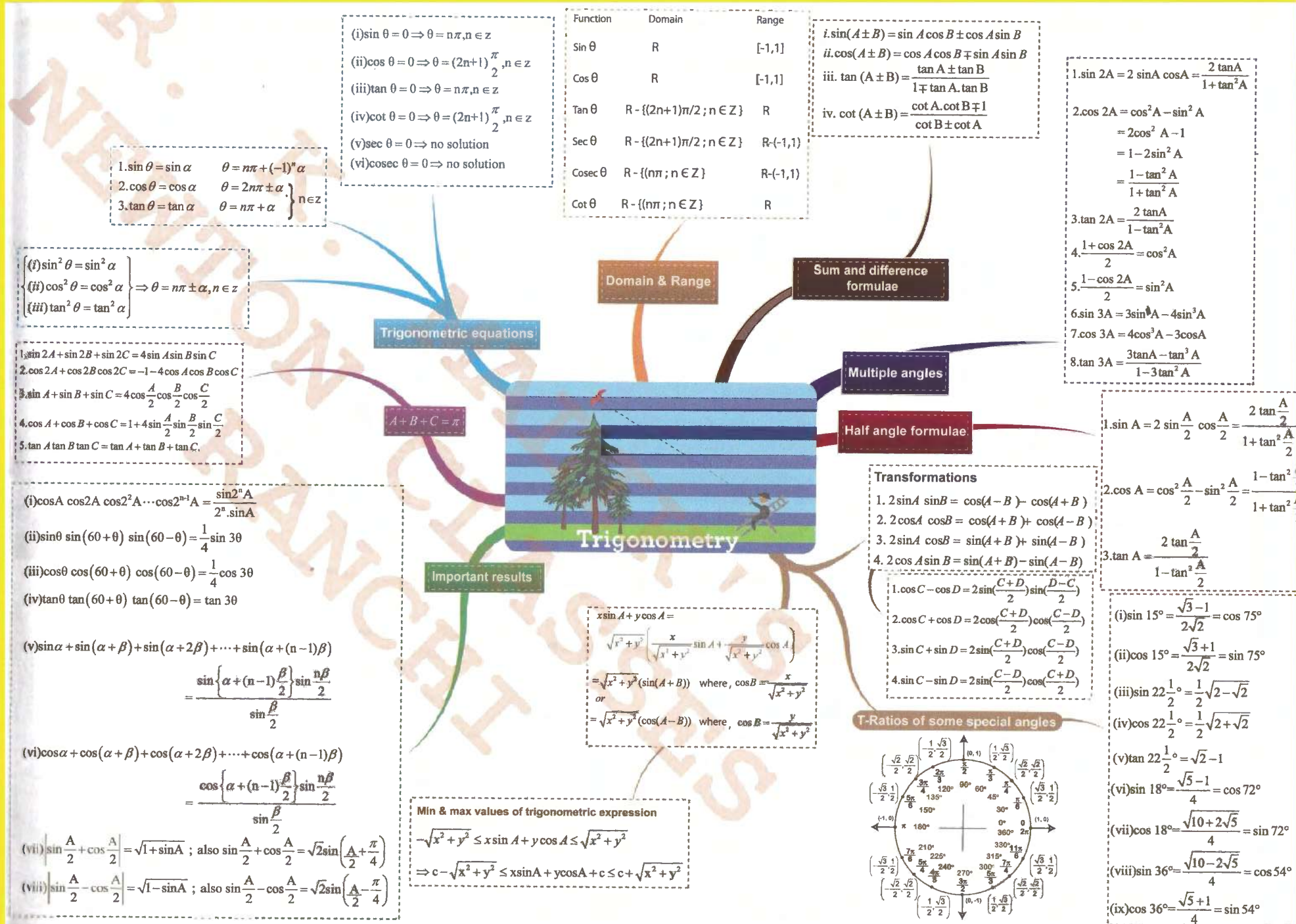
$$\begin{aligned} \text{Semi perimeter} &= s = \frac{a+b+c+d}{2} \\ \text{Area} &= \frac{1}{2} (ab + cd) \sin B \\ &= \sqrt{(s-a)(s-b)(s-c)(s-d)} \\ \cos B &= \frac{a^2 + b^2 - c^2 - d^2}{2(ab + cd)} \end{aligned}$$

Ptolemy's theorem:
In a cyclic quadrilateral ABCD
AC BD = AB CD + BC AD

Making a triangle out of any three A, B, C, & H, the fourth vertex is the orthocentre of the triangle

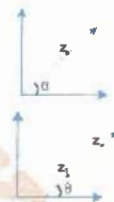
- P & H are images abt BC
- $\Delta HDC \cong \Delta PDC$
- $\Delta HBC \cong \Delta PBC$
- Perimeter of AQCPBR is $4R(\cos A + \cos B + \cos C)$ MAX $\rightarrow 6R$

- Angles $\rightarrow \pi - 2A, \pi - 2B, \pi - 2C$
- H is incentre of ΔDEF
- $EF = a \cos A, FD = b \cos B, DE = c \cos C$
- $\Delta DEF = 2\Delta \cos A \cos B \cos C$
- Radius of circumcircle of $\Delta DEF = \frac{R}{2}$
- Radius of incircle of pedal $\Delta DEF = 2R \cos A \cos B \cos C$



Some standard loci:

1) If z is a variable point argand plane such that $\arg(z) = \theta$, then locus of z is a straight line (excluding origin) through origin inclined at an angle θ



2) If z is a variable point and z_1 is a fixed point $\arg(z-z_1) = \theta$, locus of z is a straight line through z_1 (excluding z_1) and inclined at an angle θ with x axis



3) If z is a variable point and z_1, z_2 are two fixed points,

i) If $|z-z_1| = |z-z_2|$

⇒ locus of z is a $\perp$ bisector of the line segment joining z_1 and z_2

ii) $|z-z_1| + |z-z_2| = |z_1-z_2|$

⇒ locus of z is a straight line segment joining z_1 and z_2

iii) $|z-z_1| - |z-z_2| = |z_1-z_2|$

⇒ locus of z is a straight line segment joining z_1 and z_2

but z does not lie between z_1 and z_2

iv) $|z-z_1| + |z-z_2| = \text{constant} \neq |z_1-z_2|$

⇒ locus of z is an ellipse

v) $|z-z_1| - |z-z_2| = \text{constant} \neq |z_1-z_2|$

⇒ locus of z is a hyperbola.

vi) If $\left| \frac{z-z_1}{z-z_2} \right| = k$,

$k=1 \Rightarrow$ locus of z is a straight line

$k \neq 1 \Rightarrow$ locus of z is a circle

vii) $|z-z_1|^2 + |z-z_2|^2 = |z_1-z_2|^2$

⇒ locus of z is a circle with z_1 and z_2 as end points of diameter.

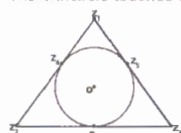
viii) $\arg\left(\frac{z-z_1}{z-z_2}\right) = \alpha (\text{fixed}) \Rightarrow$ locus of z is a segment of circle

$\arg\left(\frac{z-z_1}{z-z_2}\right) = \pm \frac{\pi}{2} \Rightarrow$ locus of z is a circle with z_1 and z_2 as end points of diameter

$\arg\left(\frac{z-z_1}{z-z_2}\right) = 0 \text{ or } \pi \Rightarrow$ locus of z is a straight line passing through z_1 and z_2

If incentre at O, z_1, z_2, z_3 are points where incircle touches sides;

z_4 on z_1z_2 ; z_5 on z_1z_3 ; z_6 on z_2z_3 ;



a) $z_1 = \frac{2z_2z_3}{z_2+z_3}$ b) $z_2 = \frac{2z_1z_3}{z_1+z_3}$ c) $z_3 = \frac{2z_1z_2}{z_1+z_2}$

1. z_1, z_2 as ends of diameter

$$(z-z_1)(\bar{z}-\bar{z}_2) + (z-z_2)(\bar{z}-\bar{z}_1) = 0$$

2. Concyclic points

$$\frac{(z_1-z_2)(\bar{z}_3-\bar{z}_4)}{(z_3-z_1)(\bar{z}_2-\bar{z}_4)} \text{ is purely real}$$

3. a. Tangent at z_1

$$\bar{z}z_1 + \bar{z}_1z = 2|z_1|^2 = 2r^2; r = \text{radius}$$

$$(\text{or}) \frac{z}{z_1} + \frac{\bar{z}}{\bar{z}_1} = 2$$

$$(\text{or}) \operatorname{Re}\left(\frac{z}{z_1}\right) = 1$$

$$|z-z_0| = R, \text{ where } z_0 \text{ is centre and } R \text{ is radius}$$

$$\bar{z}z + a\bar{z} + \bar{a}z + b = 0, \text{ whose centre is } -a \text{ and radius } R \text{ is } \sqrt{a\bar{a}-b}$$

Let $A(z_1), B(z_2), C(z_3)$ be any three points in the argand plane.

1) ABC is equilateral if and only if

$$a) z_1^2 + z_2^2 + z_3^2 = z_1z_2 + z_2z_3 + z_3z_1$$

$$b) \frac{1}{z_1-z_2} + \frac{1}{z_2-z_3} + \frac{1}{z_3-z_1} = 0$$

2) If ABC is right angled isosceles triangle with $\angle C = \frac{\pi}{2}$, then

$$(z_1 - z_2)^2 = 2(z_1 - z_3)(z_3 - z_2)$$

3) If the altitude from A to side BC meets the circumcircle

$$\text{centered at origin at } D(z_4) \text{ then } z_4 = \frac{-z_2z_3}{z_1}$$

4) If the internal bisector of angle A meets the circumcircle at $E(z_5)$ then $z_5^2 = z_2z_3$

1. Parametric form

$$z = tz_1 + (1-t)z_2 \text{ where } t \in R$$

2. Passing through

z_1, z_2 (non parametric form)

$$\begin{vmatrix} z & \bar{z} & 1 \\ z_1 & \bar{z}_1 & 1 \\ z_2 & \bar{z}_2 & 1 \end{vmatrix} = 0$$

$$3. az + \bar{a}z + b = 0 \quad a, z \in C, b \in R$$

$$\text{Its slope is } \frac{-a}{\bar{a}}$$

4. Slope of a line passing through

$$z_1 \text{ and } z_2 \text{ is } \frac{z_2 - z_1}{\bar{z}_2 - \bar{z}_1}$$

$$1. \text{Mid-point Formula } \frac{z_1 + z_2}{2}$$

$$2. \text{Section Formula } \frac{mz_1 + nz_2}{m+n}$$

$$3. \text{Centroid formula } \frac{z_1 + z_2 + z_3}{3}$$

4. z_1, z_2, z_3 are collinear
If $lz_1 + mz_2 + nz_3 = 0$,
 $l+m+n=0, l, m, n \in R$
and not all are zeroes

Line

Circle

Geometrical
Applications

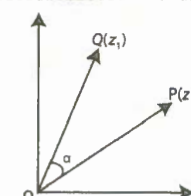
Distance:

$$d = |z_2 - z_1|$$

Distance of z_2 from z_1

Propositions

Rotation



If $\vec{OP}$ is rotated through an angle α in anti clock wise direction then $z_1 = z e^{i\alpha}$

If z_1, z_2, z_3 are three points, and if α is in anti-clockwise sense,

$$\frac{z_3 - z_1}{z_2 - z_1} = \frac{z_3 - z_1}{z_2 - z_1} e^{i\alpha}$$

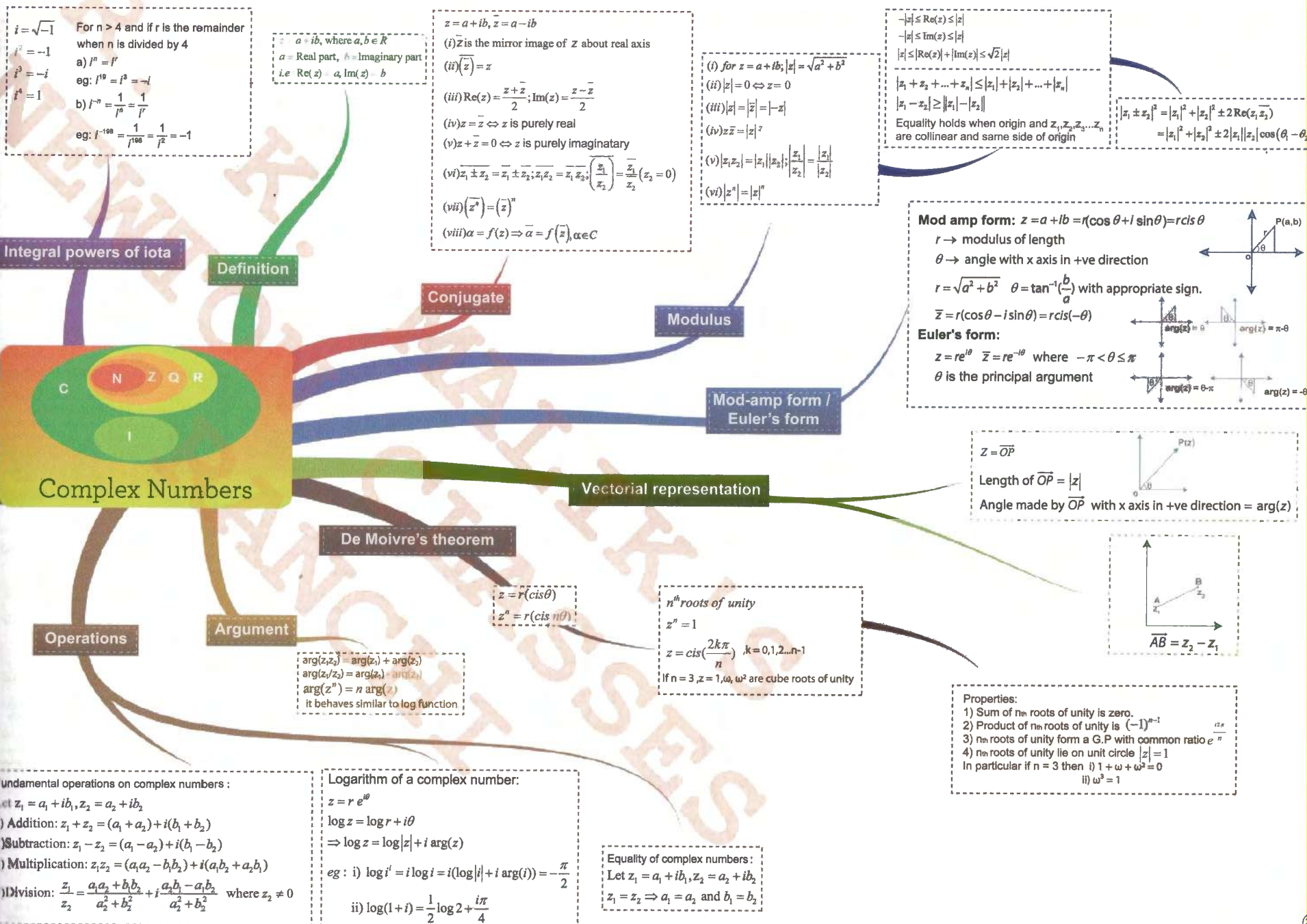


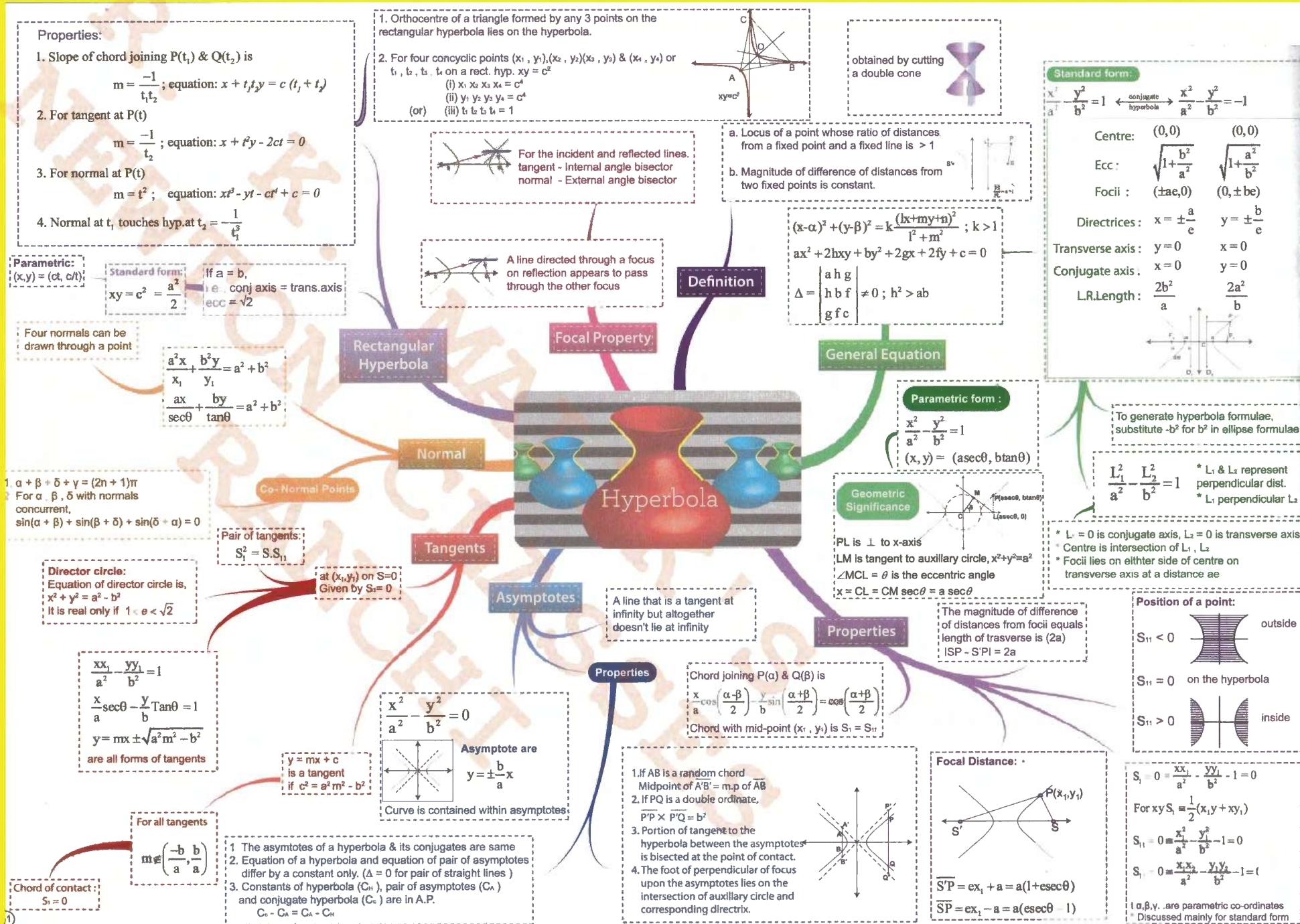
Rotate denominator line to numerator line

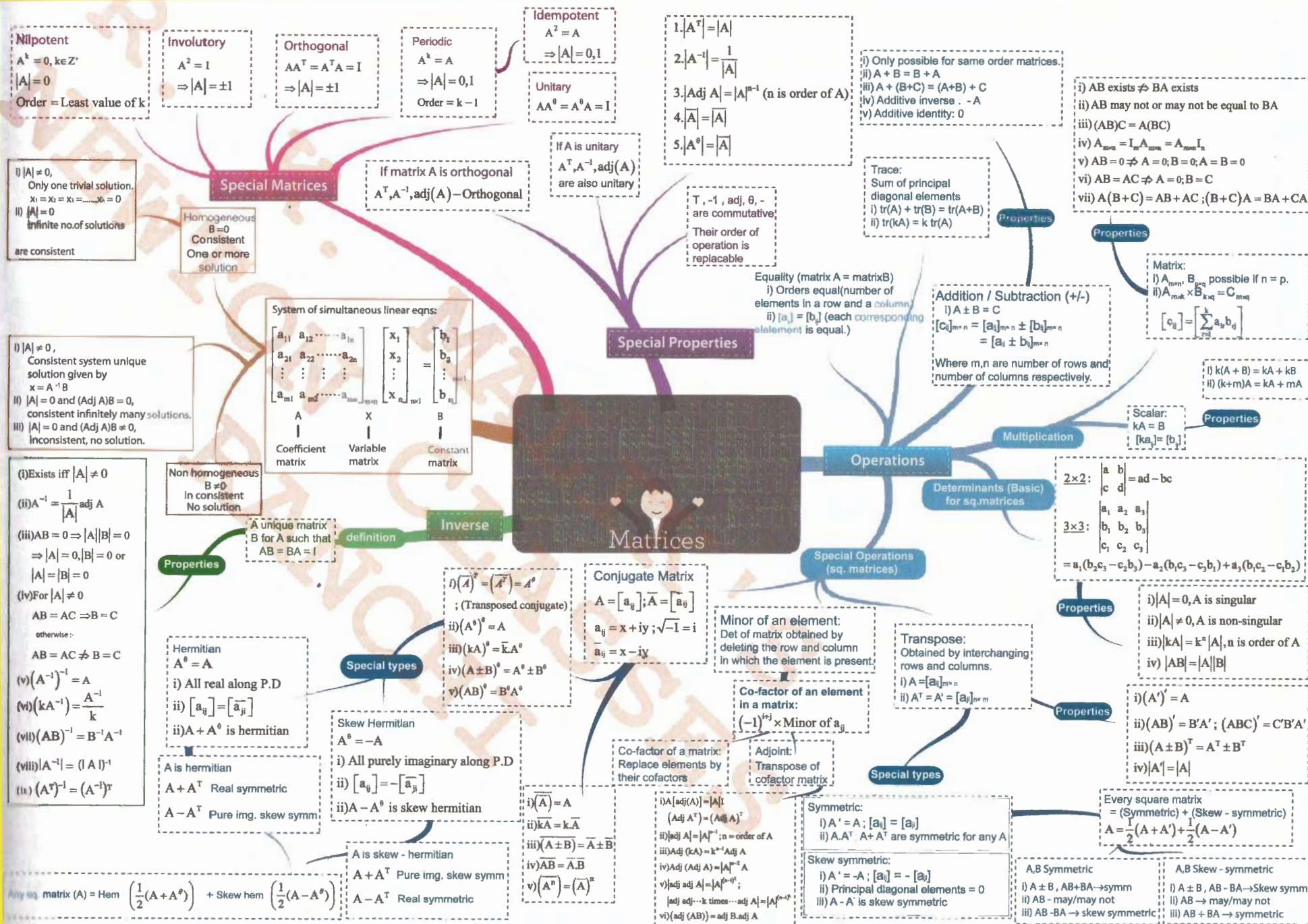
(If α is clockwise, take it as -ve)

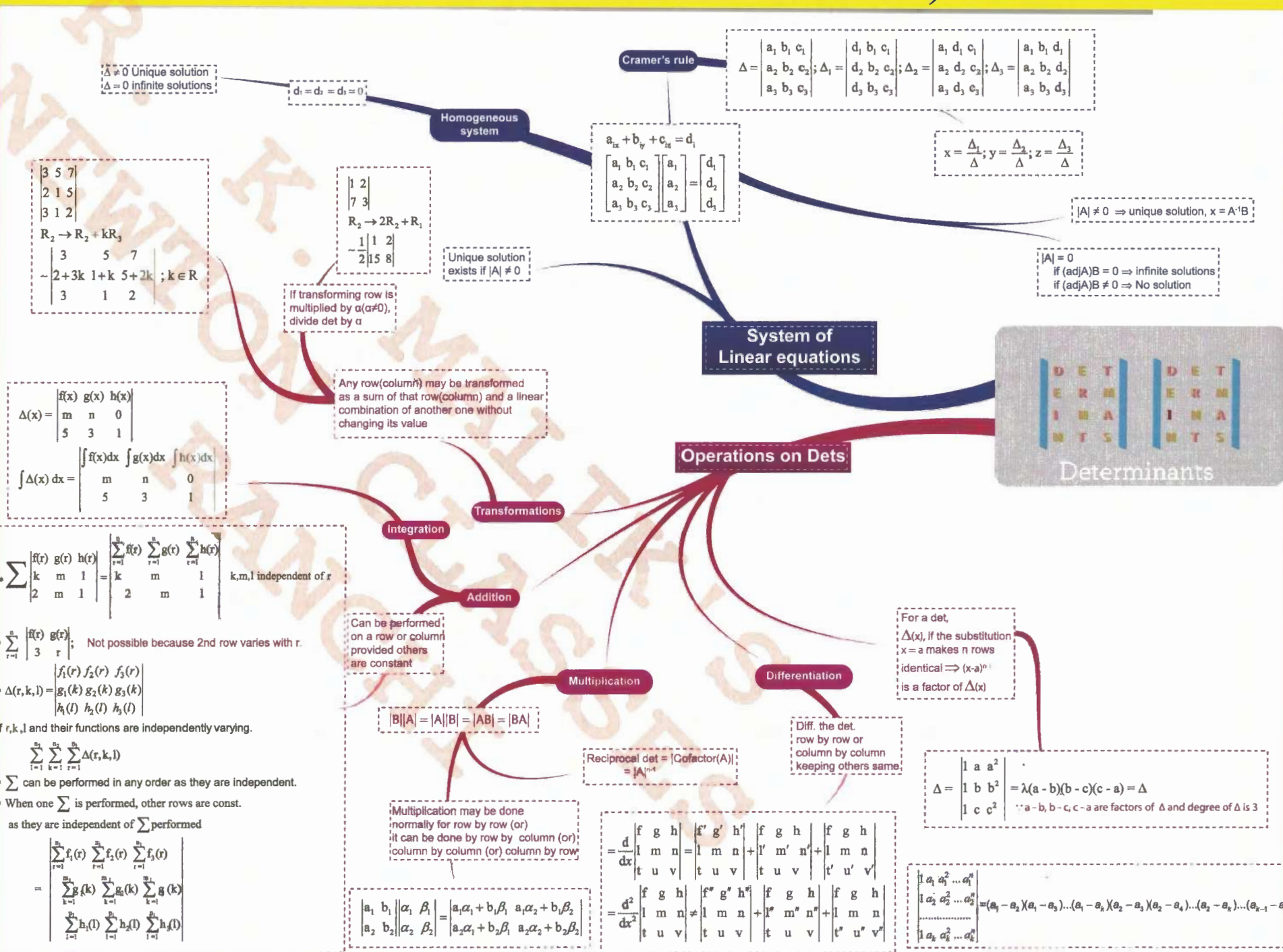
If $z = \text{Origin}$

$$\frac{z_3}{z_2} = \left| \frac{z_3}{z_2} \right| e^{i\alpha}$$









Properties

1. $|A| = |A^T|$

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

eg., $\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} = -2$; $\begin{vmatrix} 1 & 3 \\ 2 & 4 \end{vmatrix} = -2$

2. On interchanging any two rows/columns the value of the determinant changes by '-' sign

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = - \begin{vmatrix} a_1 & b_1 & c_1 \\ a_3 & b_3 & c_3 \\ a_2 & b_2 & c_2 \end{vmatrix} = - \begin{vmatrix} a_3 & b_3 & c_3 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = - \begin{vmatrix} a_2 & b_2 & c_2 \\ a_1 & b_1 & c_1 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

3. If any two rows/columns are identical or proportional, the determinant is zero

$$\begin{vmatrix} 1 & 2 & 4 \\ 3 & 1 & 1 \\ 1 & 2 & 4 \end{vmatrix} = 0 \quad \begin{vmatrix} 1 & 9 & 2 \\ 3 & 4 & 6 \\ 4 & 3 & 8 \end{vmatrix} = 0$$

4. If a row/column contains only zeroes, det

$$\begin{vmatrix} 0 & 0 & 0 & 0 \\ 24 & 16 & 78 & 931 \\ 361 & 282 & 196 & 39 \\ 27 & 927 & 641 & 789 \end{vmatrix} = 0$$

5. If all the elements of a row or column are multiplied by a constant the value of the det becomes k times

a) $k \begin{vmatrix} 1 & 3 & 7 \\ 2 & 4 & 5 \\ 3 & 1 & 2 \end{vmatrix} = \begin{vmatrix} k & 3 & 7 \\ 2k & 4 & 5 \\ 3k & 1 & 2 \end{vmatrix} = \begin{vmatrix} k & 3k & 7k \\ 2 & 4 & 5 \\ 3 & 1 & 2 \end{vmatrix} = \begin{vmatrix} 1 & 3 & 7 \\ 2 & 4 & 5 \\ 3k & k & 2k \end{vmatrix}$

b) $\begin{vmatrix} k & 2k & 3k \\ 4k & 2k & k \\ 6k & 3k & k \end{vmatrix} = k^3 \begin{vmatrix} 1 & 2 & 3 \\ 4 & 2 & 1 \\ 6 & 3 & 1 \end{vmatrix}$

6. If the elements of any row/ column are a sum of m elements the det can be written as a sum of m dets
* only if other rows are kept constant.

$$\begin{vmatrix} \alpha+3 & \beta+2 \\ 2 & 1 \end{vmatrix} = \begin{vmatrix} \alpha & \beta \\ 2 & 1 \end{vmatrix} + \begin{vmatrix} 3 & 2 \\ 2 & 1 \end{vmatrix} = \begin{vmatrix} \alpha & 2 \\ 2 & 1 \end{vmatrix} + \begin{vmatrix} 3 & \beta \\ 2 & 1 \end{vmatrix}$$

$$= \begin{vmatrix} \alpha+3 & \beta \\ 2 & 1 \end{vmatrix} + \begin{vmatrix} 0 & 2 \\ 2 & 1 \end{vmatrix}$$

7. If det is upper triangular or lower triangular, det equals product of principal diagonal elements

$$\begin{vmatrix} 1 & 3 & 2 \\ 0 & 6 & 5 \\ 0 & 0 & 3 \end{vmatrix} = 1 \times 6 \times 3 = 18; \quad \begin{vmatrix} 1 & 0 & 0 \\ 3 & 9 & 0 \\ 1 & 5 & 4 \end{vmatrix} = 1 \times 9 \times 4 = 36$$

8. If A is skew-symmetric matrix of odd order, $|A| = 0$

Definition

Determinant is obtained by adding the products of the elements of a row or column with their co-factors

$$\begin{vmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots \\ a_{n1} & \dots & \dots & a_{nn} \end{vmatrix}$$

$$= a_{11} \times c_{a11} + a_{12} \times c_{a12} + \dots + a_{1n} \times c_{a1n}$$

$$= a_{1j} \times c_{a1j} + a_{2j} \times c_{a2j} + \dots + a_{nj} \times c_{anj}$$

Common Determinants :

$$|A_{1 \times 1}| = |a_{11}| = a_{11}; \quad |A_{2 \times 2}| = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc; \quad |A_{3 \times 3}| = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

$$= a_1(b_2c_3 - b_3c_2) - b_1(a_2c_3 - a_3c_2) + c_1(a_2b_3 - a_3b_2)$$

DETERMINANTS

Determinants

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Applications:
Conditions for collinearity:
(i) A, B, C are collinear
 $\Leftrightarrow$ (i) $\Delta ABC = 0$
(ii) $AB + BC = AC$
(iii) slope of AB = slope of BC
= slope of AC

Area of a polygon with vertices
 $(x_1, y_1), (x_2, y_2), (x_3, y_3), \dots, (x_n, y_n)$ is
$$\frac{1}{2} \left| \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \\ \vdots & \vdots & \vdots \\ x_n & y_n & 1 \end{vmatrix} \right|$$

Area of a triangle with vertices
 $(x_1, y_1), (x_2, y_2), (x_3, y_3)$ is
$$\frac{1}{2} \left| \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} \right|$$

Area of a triangle with vertices
 $(x_1, y_1), (x_2, y_2), (x_3, y_3)$
$$= \frac{1}{2} \sum x_i(y_j - y_k)$$

$$= \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

$$= \frac{1}{2} \left| \begin{vmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{vmatrix} \right|$$

Area of a quadrilateral with vertices
 $(x_1, y_1), (x_2, y_2), (x_3, y_3), (x_4, y_4)$
$$= \frac{1}{2} \left| \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \\ x_4 & y_4 & 1 \end{vmatrix} \right|$$

Distance of $(a \cos \alpha, a \sin \alpha)$ from origin = a

Distance of the point
 (x, y) from the origin
 $= \sqrt{x^2 + y^2}$

For external Division
Coordinate of a point
 $= \left(\frac{mx_2 - nx_1}{m - n}, \frac{my_2 - ny_1}{m - n} \right)$

Harmonic Conjugate

P, Q are called
Harmonic Conjugates
to each other,
if $AP : PB = AQ : QB$

Section formula :

For internal division,

Coordinates of the point R
 $= \left(\frac{mx_2 + nx_1}{m + n}, \frac{my_2 + ny_1}{m + n} \right)$
 $m = n = 1$
mid point = $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$

$A(x_1, y_1), B(x_2, y_2), C(x_3, y_3)$
are consecutive vertices
of a parallelogram ABCD
Fourth Vertex D
 $= (x_1 + x_3 - x_2, y_1 + y_3 - y_2)$

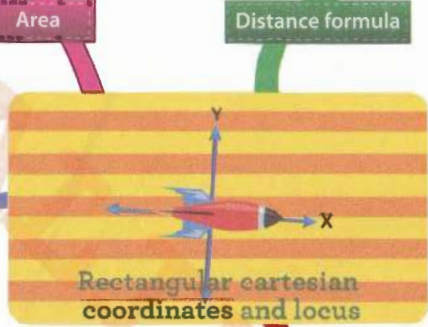
If triangle ABC is equilateral and
 $A(x_1, y_1)$ and $B(x_2, y_2)$ are any two
vertices, then 3rd vertex
 $C = \left(\frac{(x_1 + x_2) \pm \sqrt{3}(y_2 - y_1)}{2}, \frac{(y_1 + y_2) \mp \sqrt{3}(x_2 - x_1)}{2} \right)$

If $A(x_1, y_1)$ and $B(x_2, y_2)$ are the end
points of the hypotenuse of a right
angled isosceles triangle, then the
3rd vertex
 $C = \left(\frac{(x_1 + x_2) \pm (y_2 - y_1)}{2}, \frac{(y_1 + y_2) \pm (x_2 - x_1)}{2} \right)$

Ratio = $x_1 - \alpha : \alpha - x_2$
 $= y_1 - \beta : \beta - y_2$

The path traced out by a moving point
subject to certain geometric condition
(or) Set of points satisfying certain
geometrical conditions

Distance formula



Points related to triangle

Incentre is equidistant from the sides of the triangle
Incentre = $\left(\frac{ax_1 + bx_2 + cx_3}{a + b + c}, \frac{ay_1 + by_2 + cy_3}{a + b + c} \right)$

Excentres

opposite to vertex A = $\left(\frac{-ax_1 + bx_2 + cx_3}{-a + b + c}, \frac{-ay_1 + by_2 + cy_3}{-a + b + c} \right)$
opposite to vertex B = $\left(\frac{ax_1 - bx_2 + cx_3}{a - b + c}, \frac{ay_1 - by_2 + cy_3}{a - b + c} \right)$
opposite to vertex C = $\left(\frac{ax_1 + bx_2 - cx_3}{a + b - c}, \frac{ay_1 + by_2 - cy_3}{a + b - c} \right)$

Incentre of triangle formed by $(0,0), (a,0), (0,b)$ is
 $\left(\frac{ab}{a + b + \sqrt{a^2 + b^2}}, \frac{ab}{a + b + \sqrt{a^2 + b^2}} \right)$

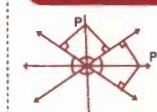
LOCUS

Definition

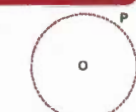
Circumcentre

(i) It is equidistant from Vertices
If S is circumcentre then $SA = SB = SC$
(ii) The circumcentre of right angled triangle is the mid point of the hypotenuse

Locus may contain finite (or) infinite points



If P is equidistant from two
intersecting lines then locus
of P is a pair of lines bisecting
the angles between them



If $OP = a$ constant,
then locus of p is a circle



If $PA = PB$, then
locus of P is
perpendicular
bisector of AB

Standard Locus

Geometrical Condition	Locus is
(i) $PA + PB = AB$	A Straight line such that A, P, B are collinear
(ii) $ PA - PB = AB$	A Straight line such that A, B, P are collinear
(iii) $PA = k \cdot PB$ ($k \neq 1$)	A circle
(iv) $PA^2 + PB^2 = AB^2$	A circle with AB as diameter
(v) $PA + PB = a$ constant ($> AB$)	An Ellipse with A, B as foci
(vi) $ PA - PB = a$ constant ($< AB$)	A Hyperbola

Orthocentre
(i) Right Angle vertex "B" is the orthocentre of right angled triangle
(ii) If O is the orthocentre of triangle ABC, then O, A, B, C are such that each point is the orthocentre of the triangle formed by the remaining three points
 $\Delta ABC \rightarrow OC$ is "O"
 $\Delta OAB \rightarrow OC$ is "C"
 $\Delta OBC \rightarrow OC$ is "A"
 $\Delta OCA \rightarrow OC$ is "B"

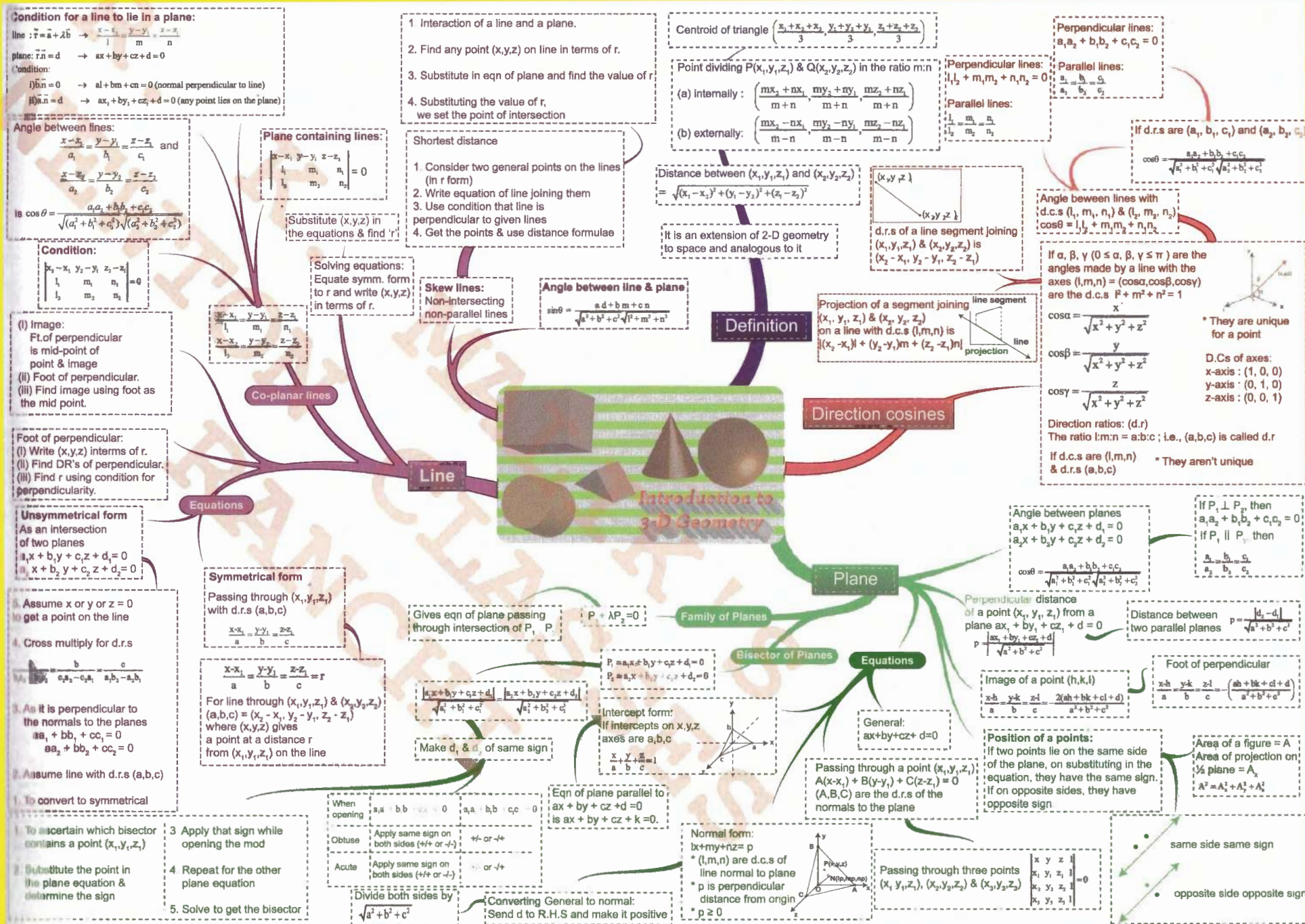
In an equilateral triangle
circumcentre, incentre,
orthocentre and centroid
coincide

Orthocentre of triangle formed by
 $(0,0), (x_1, y_1)$ and (x_2, y_2) is
 $\left(\frac{(y_2 - y_1)x_1x_2 + y_1y_2}{x_2y_1 - x_1y_2}, \frac{(x_2 - x_1)x_1x_2 + y_1y_2}{x_2y_1 - x_1y_2} \right)$

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- Two collinear or three coplanar
→ always linearly dependent.
 $\vec{a}, \vec{b}, \vec{c}$ collinear if
 $\lambda_1 \vec{a} + \lambda_2 \vec{b} + \lambda_3 \vec{c} = 0$
 $\lambda_1 + \lambda_2 + \lambda_3 = 0$
- Two non-collinear or three non-coplanar
→ always linearly independent.
- Any four vectors are linearly dependent
- $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ are coplanar if
 $\lambda_1 \vec{a} + \lambda_2 \vec{b} + \lambda_3 \vec{c} + \lambda_4 \vec{d} = 0$ & $\sum \lambda_i = 0$

Linearly dependent:
 $k_1 \vec{a}_1 + k_2 \vec{a}_2 + \dots + k_n \vec{a}_n = 0$
& not all of $k_1, k_2, \dots, k_n$ are 0
Linearly independent:
 $k_1 \vec{a}_1 + k_2 \vec{a}_2 + \dots + k_n \vec{a}_n = 0$
⇒ All of $k_1, k_2, \dots, k_n = 0$

- $\vec{a}, \vec{b}$ are two non-zero, non-collinear vectors,
any vector in the plane can be given by
 $\vec{r} = \lambda \vec{a} + \mu \vec{b}$
- $\vec{a}, \vec{b}, \vec{c}$ are any three non coplanar vectors
then any vector $\vec{r}$ in space is given by
 $\vec{r} = \lambda \vec{a} + \mu \vec{b} + \nu \vec{c}$

- Midpoint of line joining $\vec{a}$ & $\vec{b}$
 $\frac{\vec{a} + \vec{b}}{2}$
Centroid $\frac{\vec{a} + \vec{b} + \vec{c}}{3}$

- Section formula:

Internal division: $\frac{m\vec{a} + n\vec{b}}{m + n}$
external division: $\frac{m\vec{a} - n\vec{b}}{m - n}$

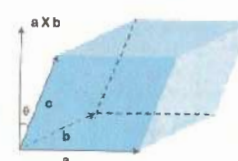
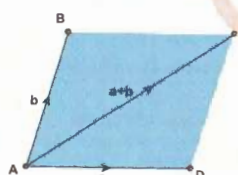
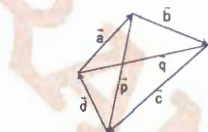
- Area of quadrilateral
with diagonals $\vec{d}_1$ & $\vec{d}_2$
 $A = \frac{1}{2} |\vec{d}_1 \times \vec{d}_2|$

- Area of triangle
with adj. sides $\vec{a}$ & $\vec{b}$
 $= \frac{1}{2} |\vec{a} \times \vec{b}|$

- Area of parallelogram
with adj. sides $\vec{a}$ & $\vec{b}$
 $= |\vec{a} \times \vec{b}|$

- Vol. of parallelepiped
with adj. sides $\vec{a}, \vec{b}, \vec{c}$
 $= |\vec{a} \cdot (\vec{b} \times \vec{c})|$

- Vol. of tetrahedron
with adj. sides $\vec{a}, \vec{b}, \vec{c}$
 $= \frac{1}{6} |\vec{a} \cdot (\vec{b} \times \vec{c})|$



Shortest distance between 2 lines

$$\vec{r} = \vec{a}_1 + \lambda \vec{b}_1 \text{ and } \vec{r} = \vec{a}_2 + \mu \vec{b}_2$$

is $\frac{(\vec{b}_1 \times \vec{b}_2) \cdot (\vec{a}_2 - \vec{a}_1)}{|\vec{b}_1 \times \vec{b}_2|}$

If the lines are parallel then

$$\text{Shortest Distance} = \frac{(\vec{a}_2 - \vec{a}_1) \times \vec{b}}{|\vec{b}|}$$

Passing through
 $\vec{a}$ and parallel to $\vec{b}$
 $\vec{r} = \vec{a} + \lambda \vec{b}$

Lines

Linear combination of vectors:
 $\vec{r} = x\vec{a} + y\vec{b} + z\vec{c} + \dots$
is a linear combination of
vectors $\vec{a}, \vec{b}, \vec{c}, \dots$ where $x, y, z, \dots$
are scalar.

Passing through
 $\vec{a}$ and $\vec{b}$
 $\vec{r} = \vec{a} + \lambda(\vec{b} - \vec{a})$

Bisector lines

$$\vec{r} = \vec{a} + \lambda \vec{b}_1$$

$$\vec{r} = \vec{a} + \mu \vec{b}_2$$

Intersect at $(\vec{a})$
⇒ $\vec{r} = \vec{a} + t(\vec{b}_1 \pm \vec{b}_2)$

$$aa' = bb' = cc' = 1$$

$$a' = \frac{\vec{b} \times \vec{c}}{[\vec{a} \vec{b} \vec{c}]}$$

$$b' = \frac{\vec{c} \times \vec{a}}{[\vec{a} \vec{b} \vec{c}]}$$

$$c' = \frac{\vec{a} \times \vec{b}}{[\vec{a} \vec{b} \vec{c}]}$$

Reciprocal
system



Vectors

Planes

Geometrical
Applications

**Collinear and non
collinear vectors**
Collinear vectors:
 $\vec{a}, \vec{b}$ are collinear
⇒ $\vec{a} = \lambda \vec{b}$ (or) $\vec{b} = \lambda \vec{a}$
for some scalar λ
Non collinear vectors
 $\vec{a}, \vec{b} \neq 0$ are non collinear
such that
 $x\vec{a} + y\vec{b} = 0 \Rightarrow x = y = 0$

Collinear points:

Points A, B, C are collinear
⇔ $\vec{AB} = \lambda \vec{BC}$
 $\vec{AB}$ (or) $\vec{AC} = \lambda \vec{BC}$
 $\vec{BC}$ (or) $\vec{AC} = \lambda \vec{AB}$

Components of vector:

in plane $\vec{OP} = x\hat{i} + y\hat{j}$ in space $\vec{OP} = x\hat{i} + y\hat{j} + z\hat{k}$
 $|\vec{OP}| = \sqrt{x^2 + y^2}$ $|\vec{OP}| = \sqrt{x^2 + y^2 + z^2}$

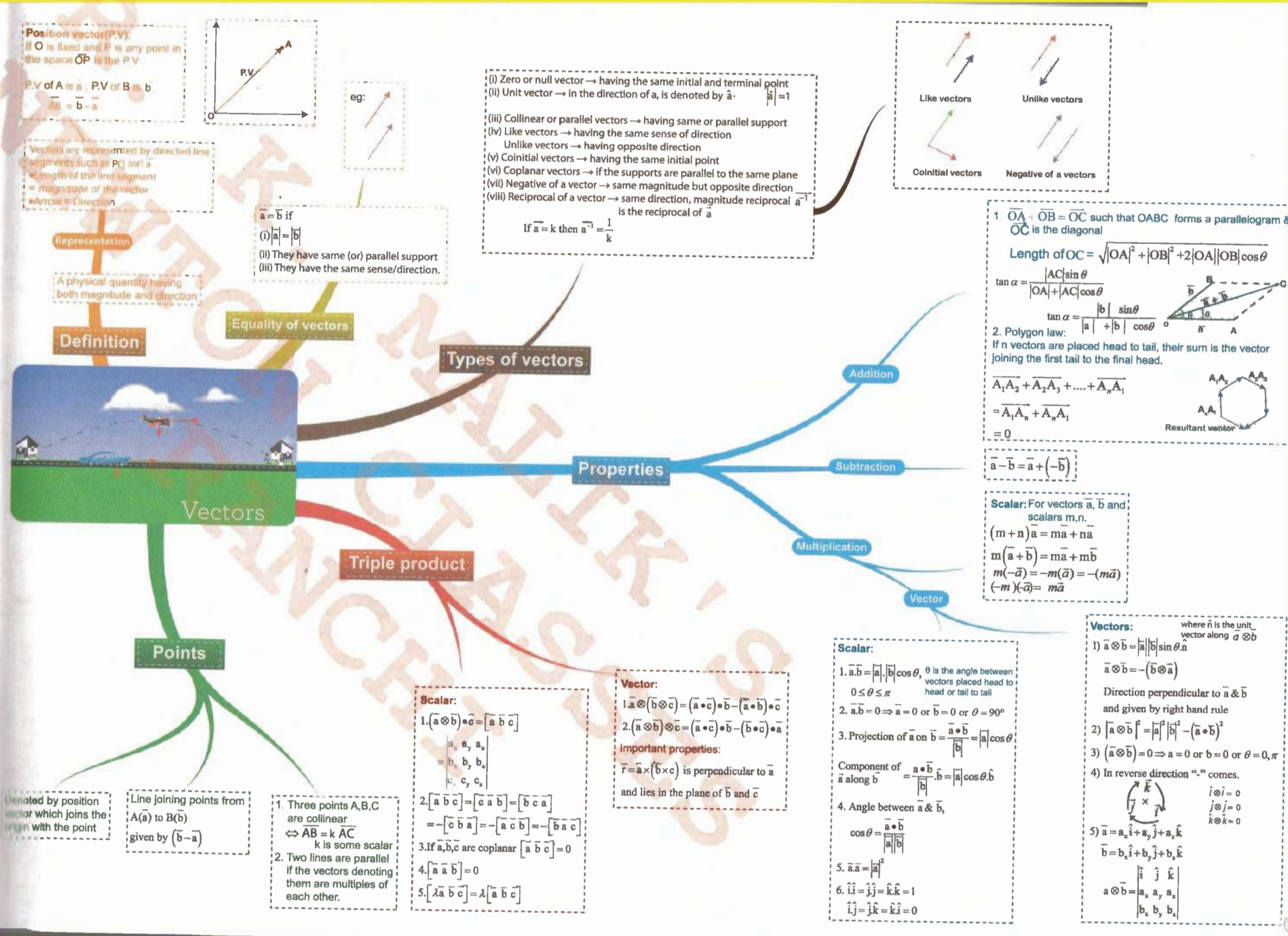
- Perpendicular form:**
Perpendicular distance
from origin = d
Normal unit vector = $\hat{n}$
 $\vec{r} \cdot \hat{n} = d$

- Passing through $\vec{a}$ and
normal to $\vec{n}$
 $(\vec{r} - \vec{a}) \cdot \vec{n} = 0$

- Passing through $\vec{a}$ and
parallel to the plane of
vectors $\vec{b}$ and $\vec{c}$
 $\vec{r} = \vec{a} + \lambda \vec{b} + \mu \vec{c}$
(or) $\vec{r} \cdot (\vec{b} \times \vec{c}) = [\vec{a} \vec{b} \vec{c}]$
ie, $[\vec{r} \vec{b} \vec{c}] = [\vec{a} \vec{b} \vec{c}]$

Passing through $\vec{a}, \vec{b}$
and parallel to $\vec{c}$
 $\vec{r} = \vec{a} + \lambda(\vec{b} - \vec{a}) + \mu \vec{c}$
 $= (1 - \lambda)\vec{a} + \lambda\vec{b} + \mu \vec{c}$

- Passing through
non-collinear $\vec{a}, \vec{b}, \vec{c}$
 $\vec{r} = (1 - \lambda - \mu)\vec{a} + \lambda\vec{b} + \mu \vec{c}$
- Angle between planes
 $\cos \theta = \hat{n}_1 \cdot \hat{n}_2$



$\frac{dy}{dx} + P(x)y = Q(x)$

Integrating Factor
 $I.F = e^{\int P(x)dx}$

Solution is
 $y.I.F = \int Q(x).I.F.d x + c$

$\frac{dx}{dy} + P(y)x = Q(y)$

$I.F = e^{\int P(y)dy}$

Solution is
 $x.I.F = \int Q(y).I.F.dy + c$

Solution is $y.e^{\int (1-n)P dx} = \int [(1-n)Q]e^{\int (1-n)P dx} dx + c$

Put $y^{1-n} = v$ and $(1-n)y^{-n} \frac{dy}{dx} = \frac{dv}{dx}$

$\therefore \frac{1}{1-n} \frac{dv}{dx} + Pv = Q$

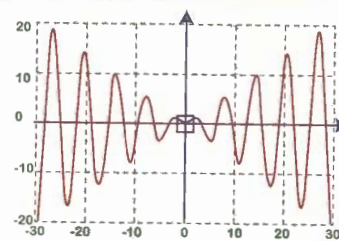
$\Rightarrow \frac{dv}{dx} + (1-n)Pv = (1-n)Q$

Dividing both sides by y^n , we get

$y^n \frac{dy}{dx} + p.y^{1-n} = Q$

$\frac{dy}{dx} + Py = Q.y^n$

where P, Q are functions of only (0,1)



An equation involving dependent variable, independent variable and derivatives of dependent variables w.r.t independent variable.

eg: (i) $\frac{dy}{dx} = xy$ (ii) $\frac{d^2 y}{dx^2} - 3 \frac{dy}{dx} + 2y = x^2$

Order: The order of the highest order derivative present in the Differential Equation. It is a positive integer.

Degree: The degree or the power of the highest order derivative in the differential equation when it is made free from fractions and radicals, i.e., when differential equation is written as a polynomial in all derivatives

eg:

Differential Equation	order	degree
(i) $\frac{d^2 y}{dx^2} + 3 \frac{dy}{dx} + y = e^x$	2	1
(ii) $y = x \frac{dy}{dx} + \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$	1	2
(iii) $\frac{d^2 y}{dx^2} = \sin\left(\frac{dy}{dx}\right)$	2	not defined

Linear Differential Equations

Inspection Method

Bernoulli's Equation

Methods of solving First Order, first degree Differential Equation

(i) Solution of $f(x, y)dx + g(x, y)dy = 0$ is $f(x, y) + g(x, y) = c$

To find the solution, the following results are useful

- (i) $x dy + y dx = d(xy)$
- (ii) $\frac{x dy + y dx}{xy} = d[\log xy]$
- (iii) $\frac{x dx + y dy}{2} = d\left[\frac{x^2 + y^2}{2}\right]$
- (iv) $\frac{2x dx + 2y dy}{x^2 + y^2} = d[\log(x^2 + y^2)]$
- (v) $\frac{x dy - y dx}{x^2} = d\left(\frac{y}{x}\right)$
- (vi) $\frac{x dy - y dx}{y^2} = -d\left(\frac{y}{x}\right)$
- (vii) $\frac{x dy - y dx}{xy} = d\left(\log \frac{y}{x}\right)$
- (viii) $\frac{x dy - y dx}{x^2 + y^2} = d(\tan^{-1} \frac{y}{x})$

Variable Seperable :

$\frac{dy}{dx} = \phi(x, y)$

$\Rightarrow f(y) dy = g(x) dx$

Solution is
 $\int f(y) dy = \int g(x) dx + c$

Solution of $\frac{dy}{dx} = f(x)$ is $y = \int f(x) dx + c$

Reducible to variable seperable

$\frac{dy}{dx} = f(ax + by + c)$

Substitute $ax + by + c = v$

Suppose $\frac{a_1}{a_2} = \frac{b_1}{b_2} = k$

substitute $a_1 x + b_1 y = v$

and $\frac{dv}{dx} = a_1 + b_1 \frac{dy}{dx}$

Reducible to Homogenous Differential Equation

(i) $\frac{dy}{dx} = \frac{a_1 x + b_1 y + c_1}{a_2 x + b_2 y + c_2}$

Put $x = X + h$
 $y = Y + k$

Where h, k is solution of $a_1 x + b_1 y + c_1 = 0$ and $a_2 x + b_2 y + c_2 = 0$

$\frac{dY}{dX} = \frac{a_1 X + b_1 Y}{a_2 X + b_2 Y}$ Substitute $Y = VX$

Solutions

Solution of Differential Equation can be found by specific standard methods

To find the solutions, integrate the Differential Equation

Homogenous Differential equation $f(Kx, Ky) = K^n f(x, y), K \neq 0$

$\frac{dy}{dx} = \frac{f(x, y)}{g(x, y)}$ is first order first degree homogenous differential equation

substitute $y = vx$

$\Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$

Formation

Formed by eliminating arbitrary constants

If the order of differential equation is n, differentiate the equation n times to eliminate arbitrary constants

Order of differential Equation = Number of arbitrary constants in Differential Equation

eg: For $y = Ae^x$, $\frac{dy}{dx} = y$ is the differential Equation

First order and first degree Differential Equation

Solution of $\frac{dy}{dx} = f(y)$ is $x = \int \frac{1}{f(y)} dy + c$

General solution: Contains as many arbitrary constants as order of Equation.

Particular : Solution obtained by assigning values to arbitrary constants

- (i) $f\left(x, y, \frac{dy}{dx}\right) = 0$
- (or)
- (ii) $\frac{dy}{dx} = f(x, y)$
- (or)
- (iii) $f(x, y)dx + g(x, y)dy = 0$
- (or)
- (iv) $\frac{dy}{dx} = \frac{f(x, y)}{g(x, y)}$
- (or)
- (v) $\frac{dy}{dx} = \phi(x, y)$

